CSCI 566: Deep Learning and its Applications

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Credits: prior USC CSCI 566 materials; Stanford CS224W.

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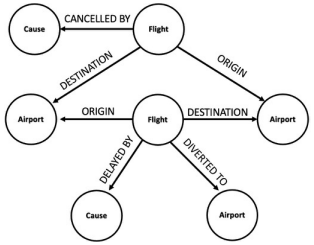
Graph Neural Networks (GNN)

Basics

Why Graphs?

Graphs are a general language for describing and analyzing entities with relations/interactions

Many Types of Data are Graphs (1)



Event Graphs

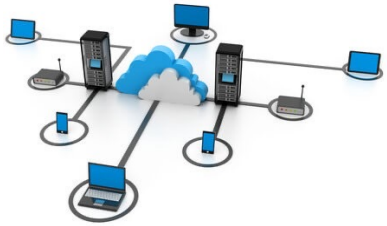
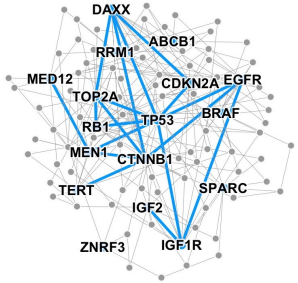


Image credit: SalientNetworks

Computer Networks



Disease Pathways

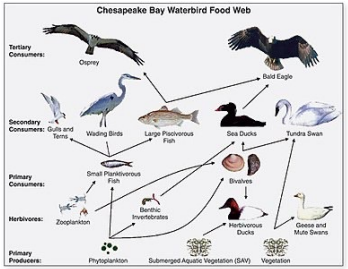


Image credit: Wikipedia

Food Webs

Image credit: Pinterest

Particle Networks

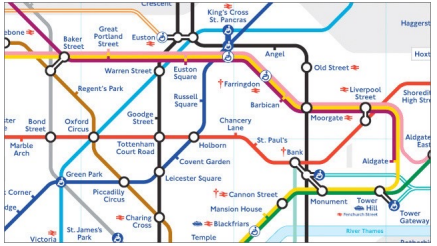


Image credit: visitlondon.com

Underground Networks

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Many Types of Data are Graphs (2)

Image credit: Medium

Social Networks

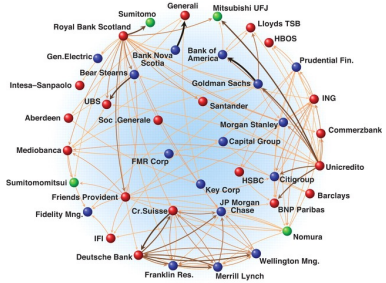
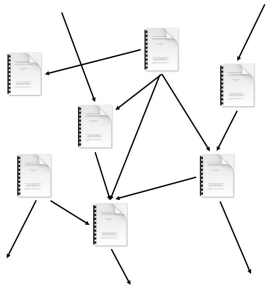


Image credit: Science

Economic Networks

Image credit: Lumen Learning

Communication Networks



Citation Networks

Image credit: Missoula Current News

Internet

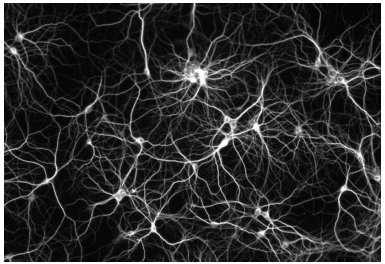


Image credit: The Conversation

Networks of Neurons

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Many Types of Data are Graphs (3)

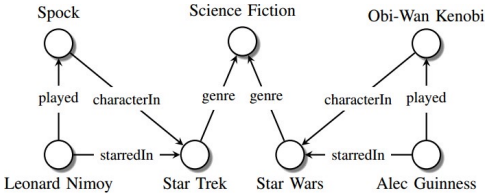


Image credit: Maximilian Nickel et al

Knowledge Graphs

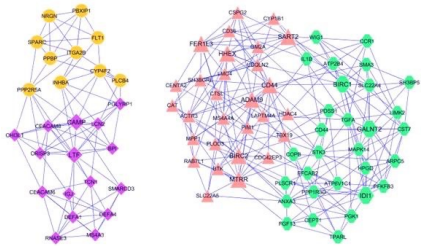


Image credit: ese.wustl.edu

Regulatory Networks

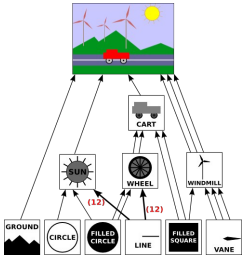


Image credit: math.hws.edu

Scene Graphs

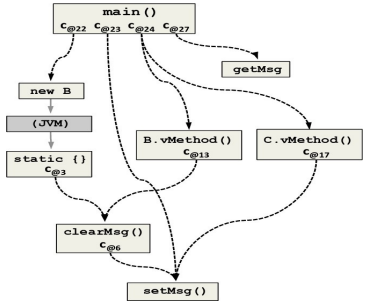
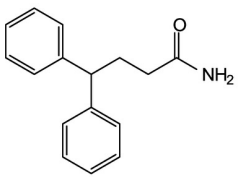


Image credit: ResearchGate

Code Graphs



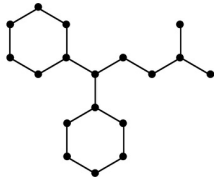


Image credit: MDPI

Molecules

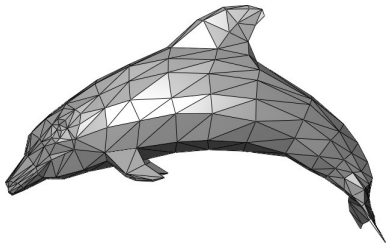


Image credit: Wikipedia

3D Shapes

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Graphs: Machine Learning

Complex domains have a rich relational structure, which can be represented as a **relational graph**

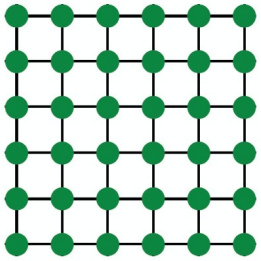
By explicitly modeling relationships, we can use this structure.

Main question:

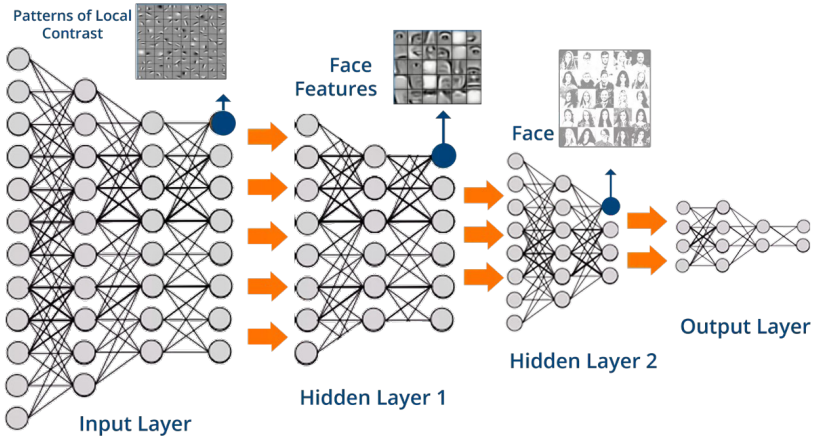
How do we take advantage of relational structure for better prediction?

Today: Modern ML Toolbox

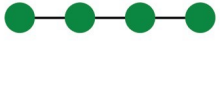
CNN

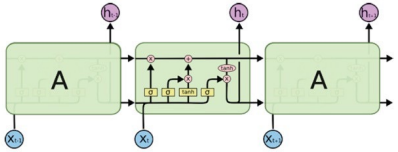


Images



Text/Speech

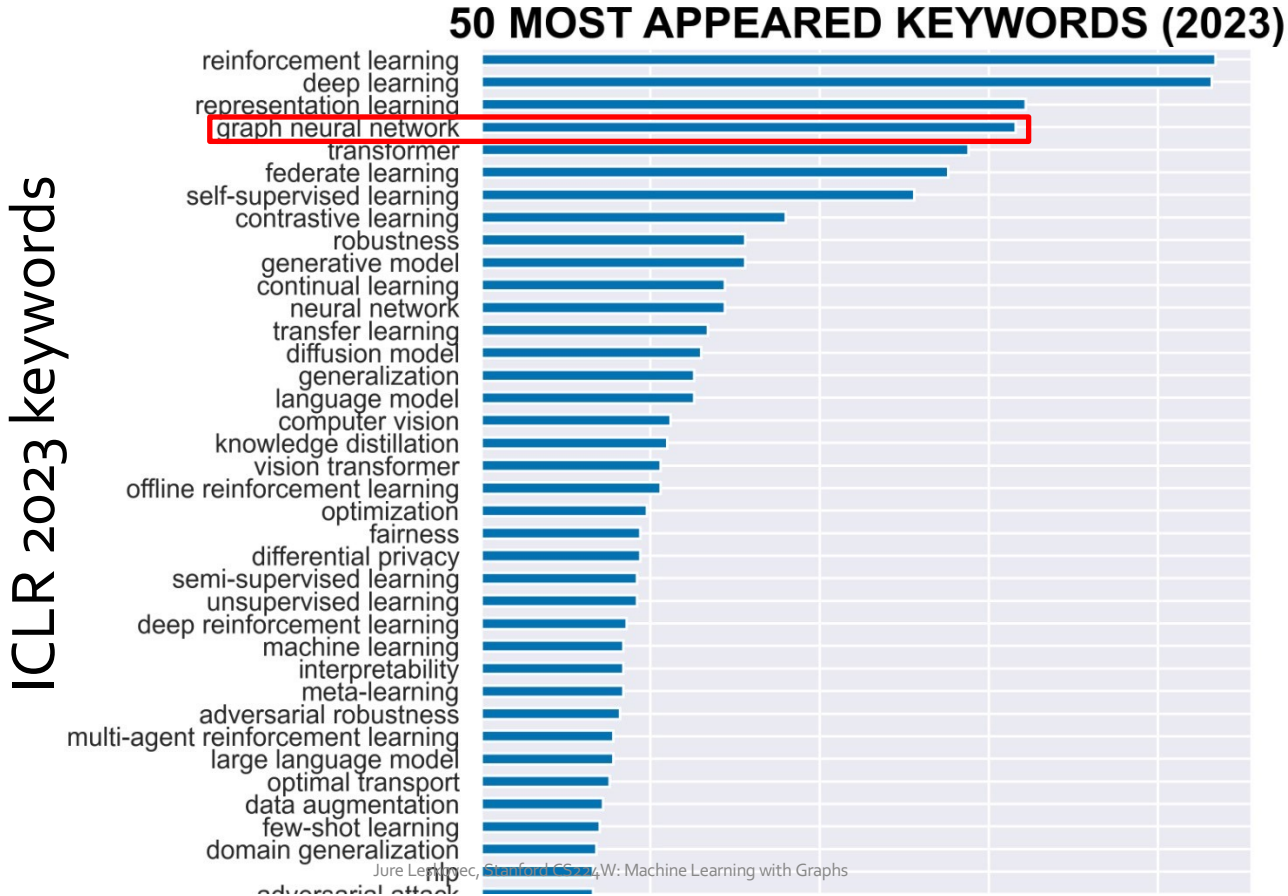




RNN/ LSTM

Modern deep learning toolbox is designed for simple sequences & grids

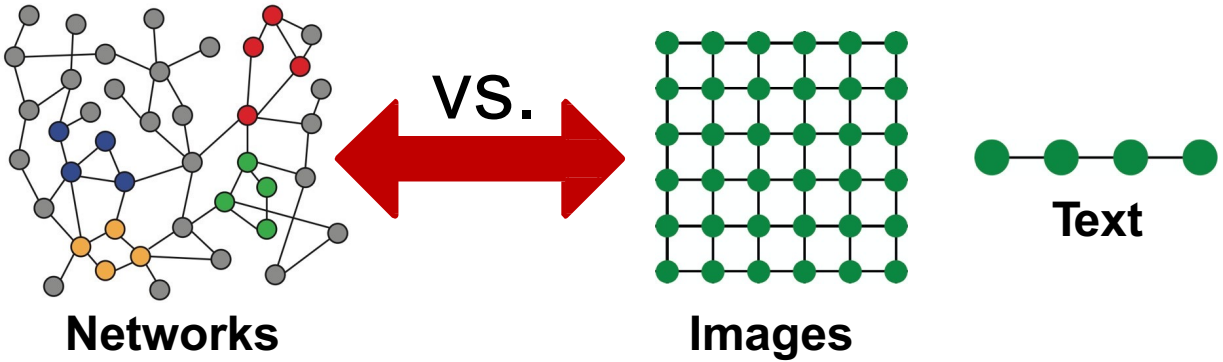
Hot subfield in ML



Why is Graph Deep Learning Hard?

Networks are complex:

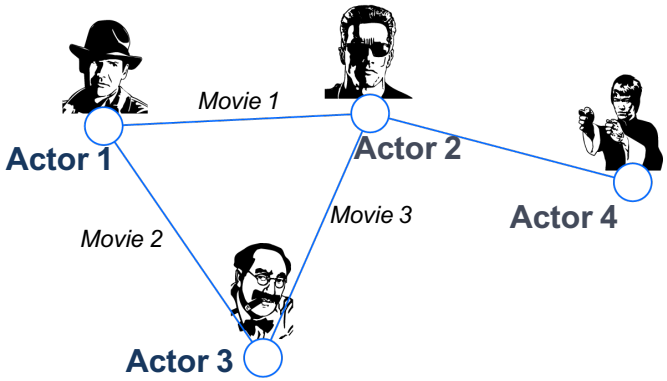
Arbitrary size and complex topological structure (*i.e.*, no spatial locality like grids)

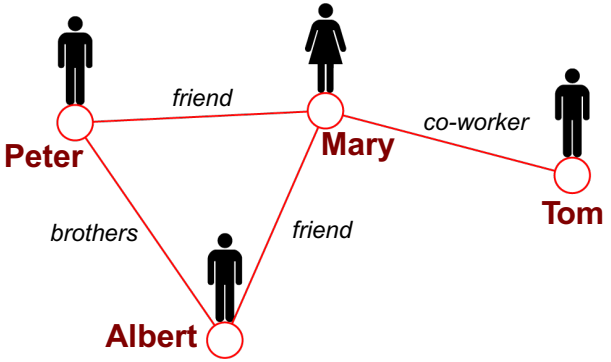


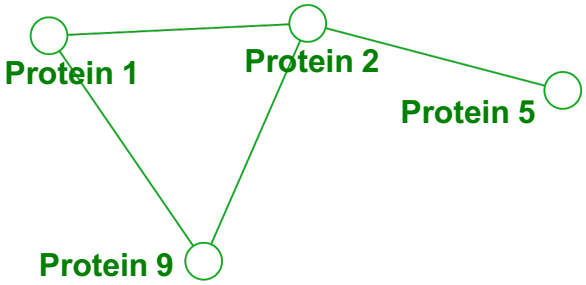
No fixed node ordering or reference point

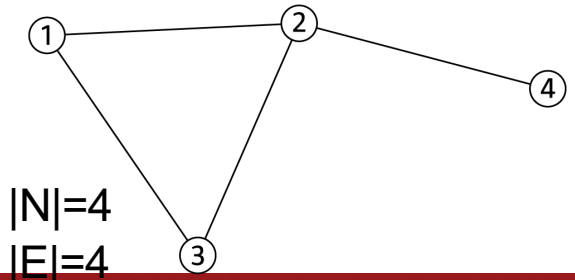
Often dynamic and have multimodal features

Graphs: A Common Language









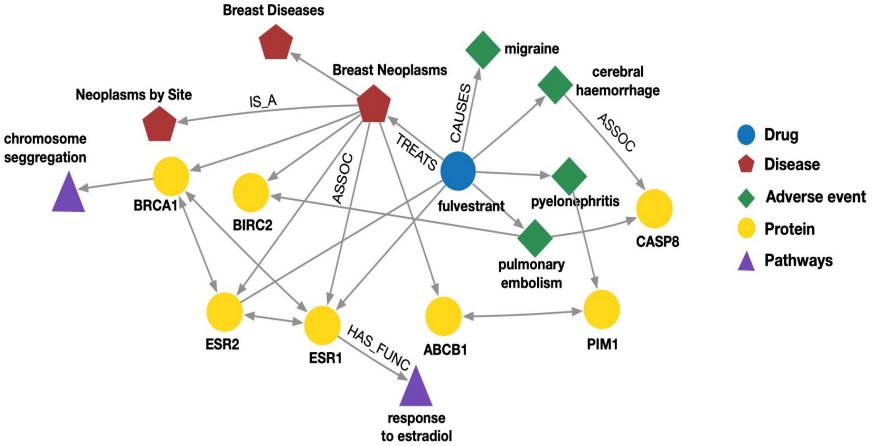
Heterogeneous Graphs

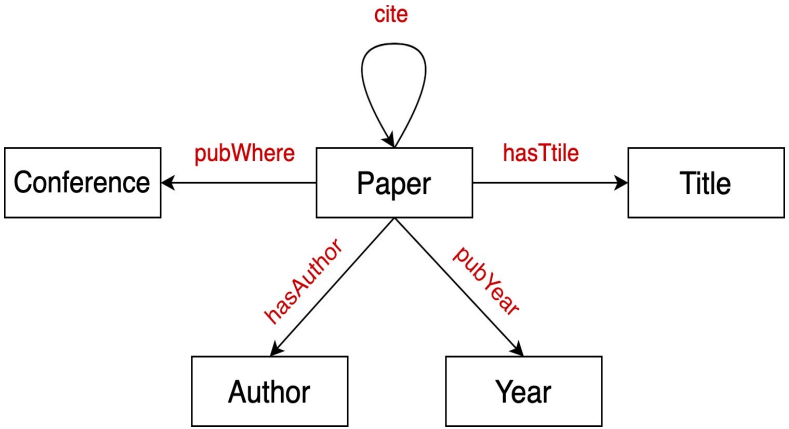
A heterogeneous graph is defined as

$$\boldsymbol{G = (V, E, R, T)}$$

- Nodes with node types $v_i \in V$
- Edges with relation types $(v_i, r, v_j) \in E$
- Node type $T(v_i)$
- Relation type $r \in R$
- Nodes and edges have attributes/features

Many Graphs are Heterogeneous





Biomedical Knowledge Graphs

Example node: Migraine
Example edge: (fulvestrant, Treats, Breast Neoplasms)
Example node type: Protein
Example edge type (relation): Causes

Academic Graphs

Example node: ICML
Example edge: (GraphSAGE, NeurIPS)
Example node type: Author
Example edge type (relation): pubYear

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Choosing a Proper Representation

How to build a graph:

- **What are nodes?**
- **What are edges?**

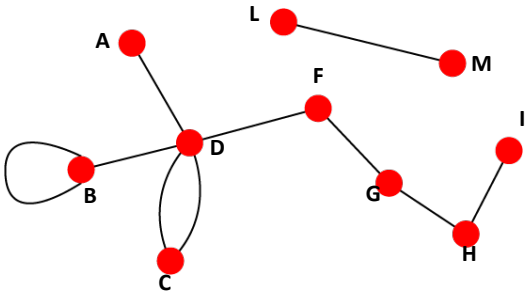
Choice of the proper network representation of a given domain/problem determines our ability to use networks successfully:

- In some cases, there is a unique, unambiguous representation
- In other cases, the representation is by no means unique
- The way you assign links will determine the nature of the question you can study

Directed vs. Undirected Graphs

Undirected

Links: undirected (symmetrical, reciprocal)

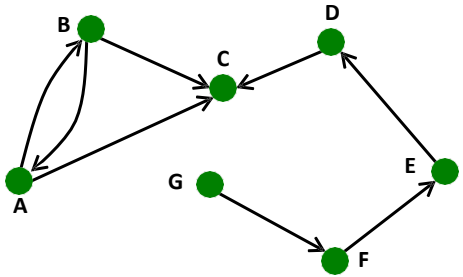


Other considerations:

- Weights
- Properties

Directed

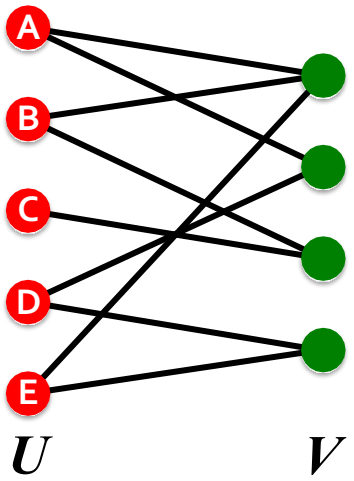
Links: directed



- Types
- Attributes

Bipartite Graph

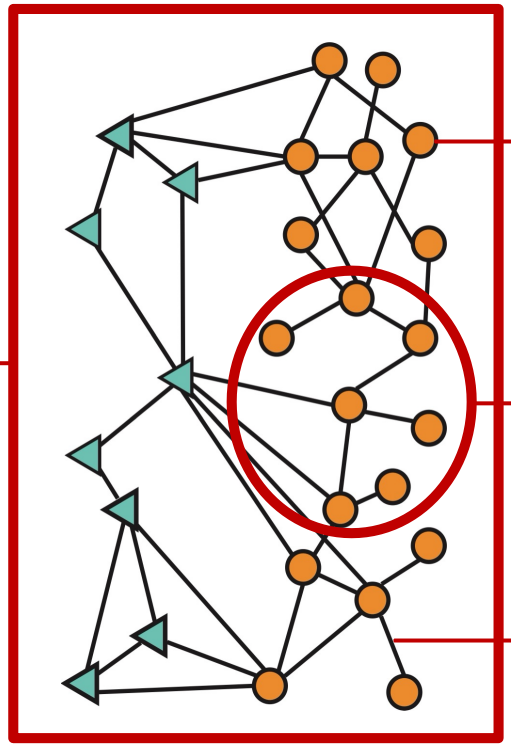
Bipartite graph is a graph whose nodes can be divided into two disjoint sets ***U*** and ***V*** such that every link connects a node in ***U*** to one in ***V***; that is, ***U*** and ***V*** are **independent sets**



Examples:

- Authors-to-Papers (they authored)
- Actors-to-Movies (they appeared in)
- Users-to-Movies (they rated)
- Recipes-to-Ingredients (they contain)

Different Types of Tasks



Graph-level prediction, Graph generation

Node level

Community (subgraph) level

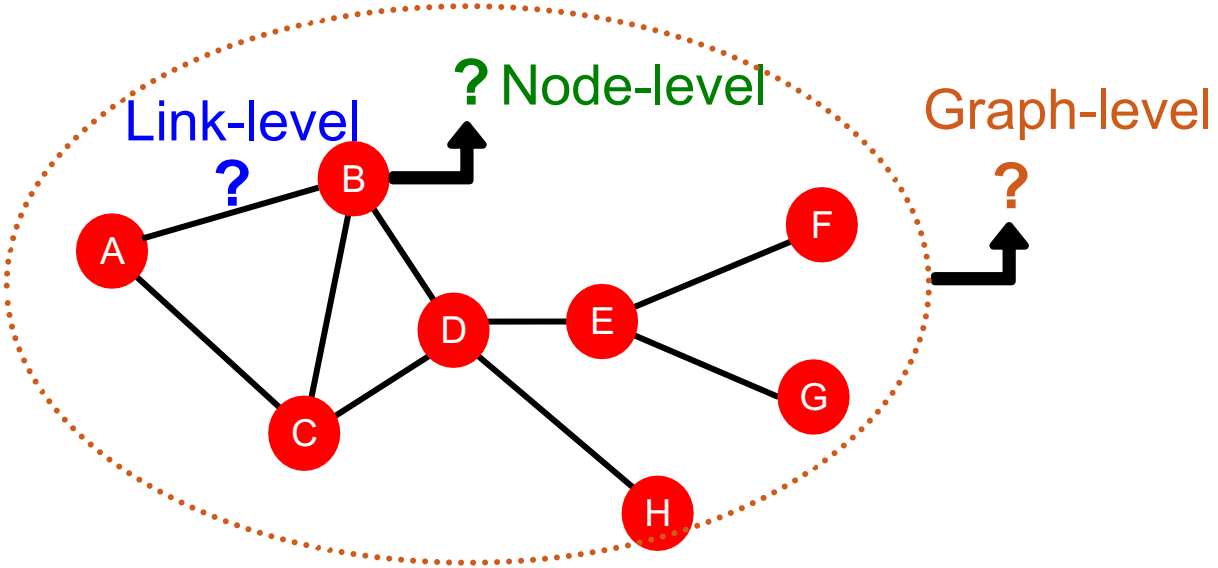
Edge-level

Machine Learning Tasks: Review

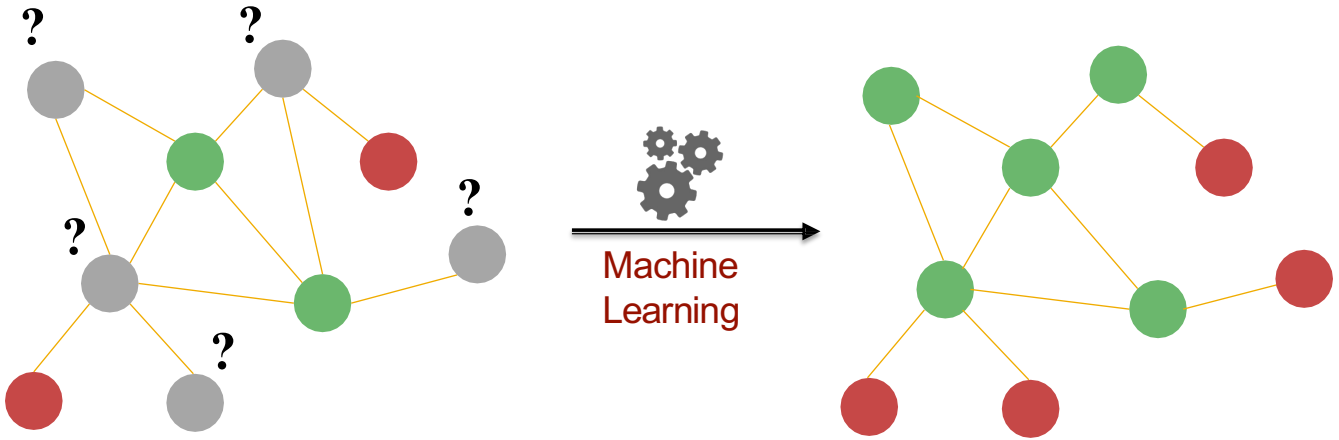
Node-level prediction

Link-level prediction

Graph-level prediction



Node-Level Tasks

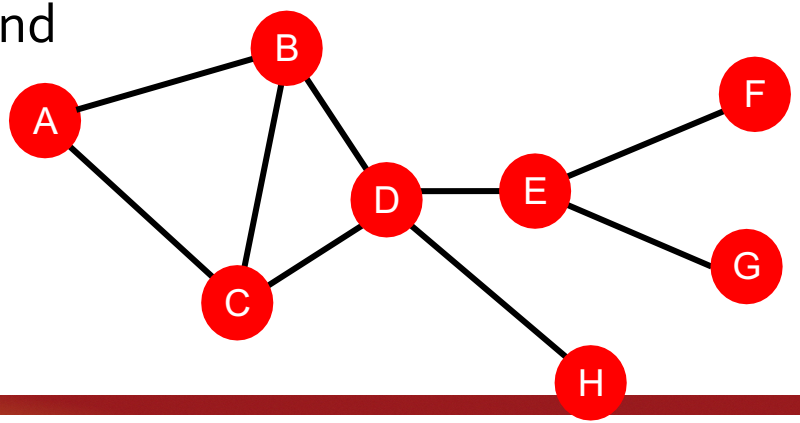


Node classification

Node-Level Network Structure

Goal: Characterize the structure and position of a node in the network:

- Node degree
- Node importance & position
  - E.g., Number of shortest paths passing through a node
  - E.g., Avg. shortest path length to other nodes
- Substructures around the node

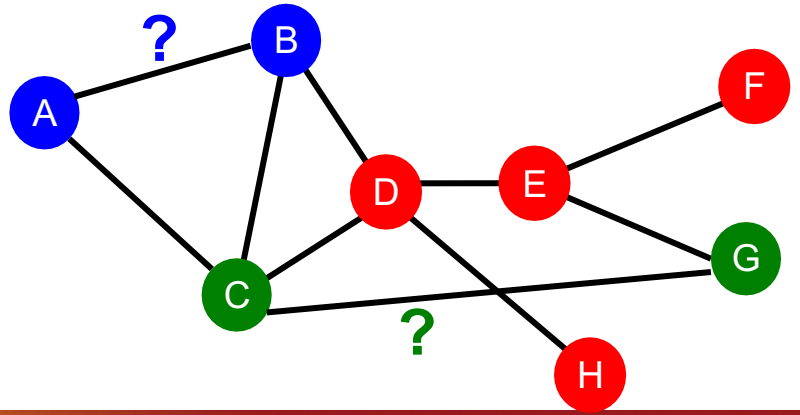


Link-Level Prediction Task

The task is to predict **new/missing/unknown links** based on the existing links.

At test time, node pairs (with no existing links) are ranked, and top K node pairs are predicted.

Task: Make a prediction for **a pair of nodes**.

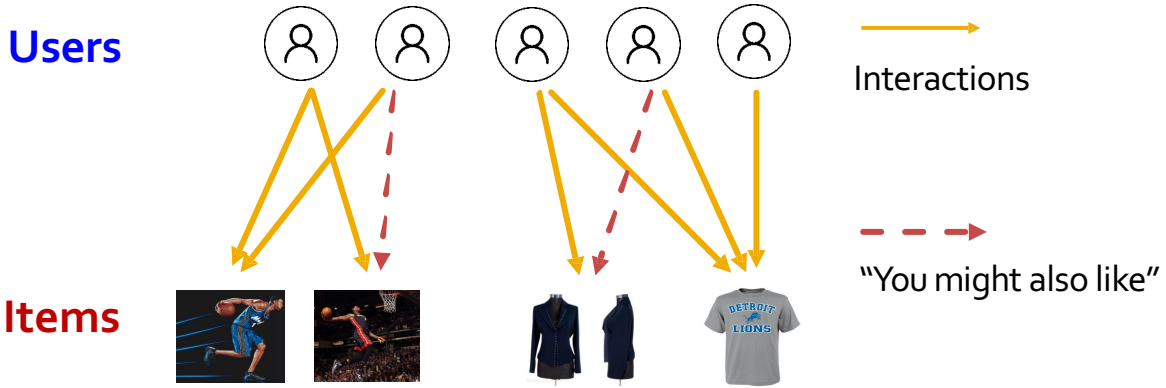


Example (1): Recommender Systems

Users interacts with items

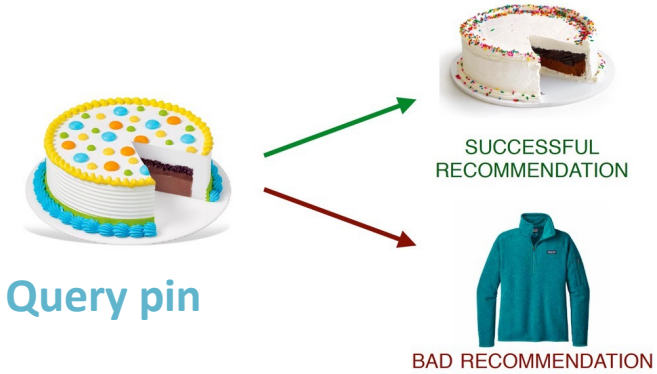
- Watch movies, buy merchandise, listen to music
- **Nodes:** Users and items
- **Edges:** User-item interactions

Goal: Recommend items users might like



PinSage: Graph-based Recommender

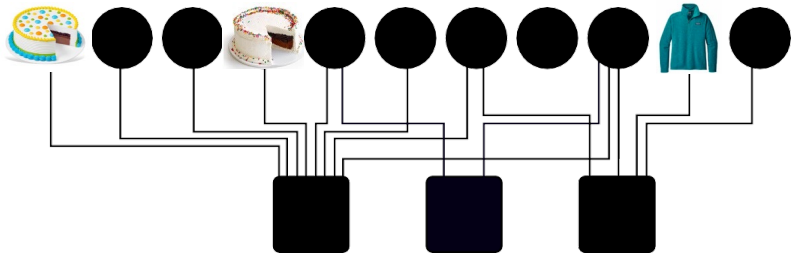
Task: Recommend related pins to users

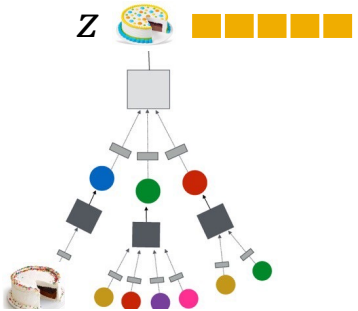


Task: Learn node embeddings z_i such that

$$d(z_{cake1}, z_{cake2}) < d(z_{cake1}, z_{sweater})$$

Predict whether two nodes in a graph are related

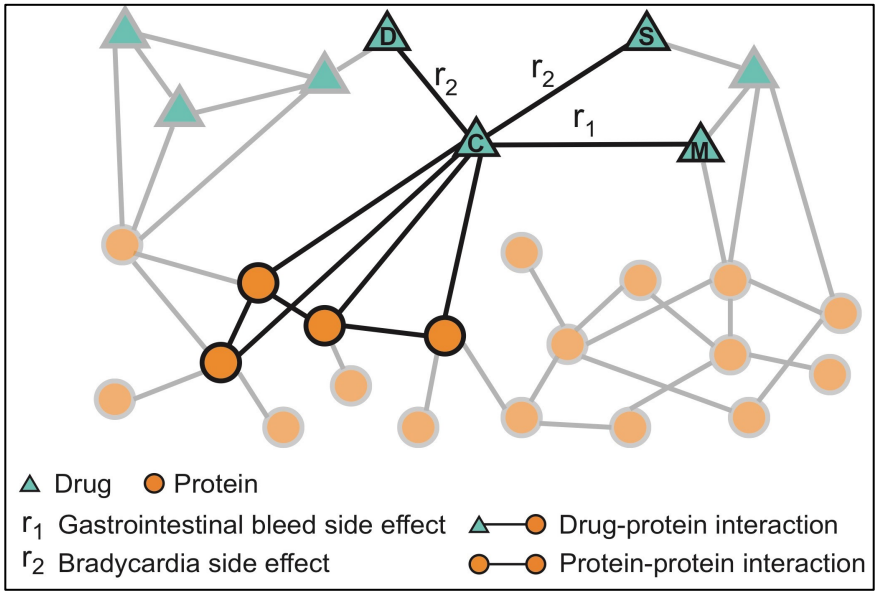




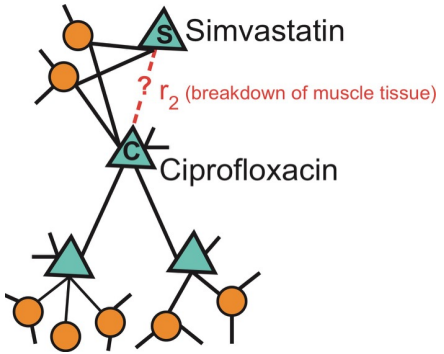
Biomedical Graph Link Prediction

Nodes: Drugs & Proteins

Edges: Interactions



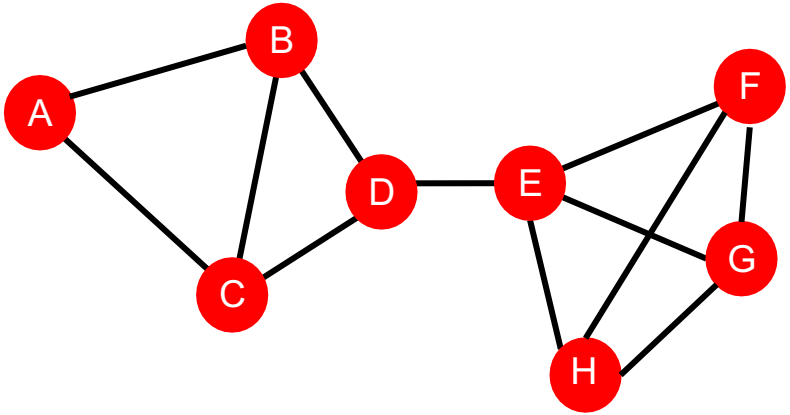
Query: How likely will Simvastatin and Ciprofloxacin, when taken together, break down muscle tissue?



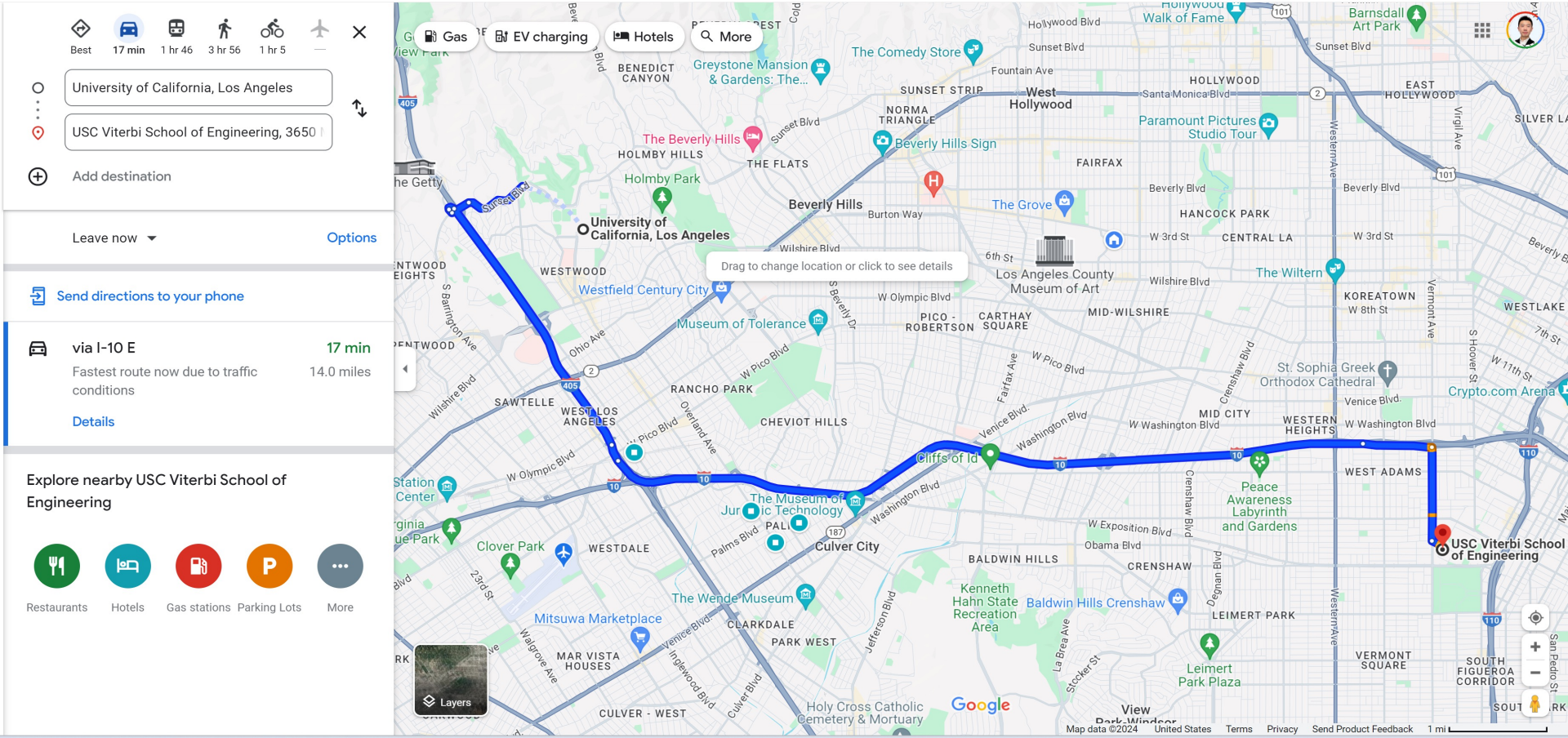
Graph-Level Features

Goal: We want make a prediction for an entire graph or a subgraph of the graph.

For example:



Example (1): Traffic Prediction

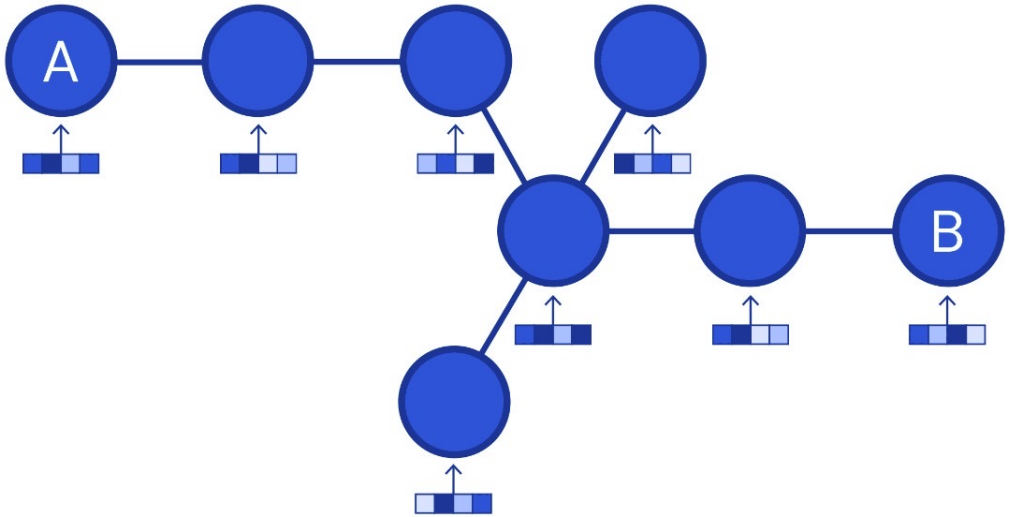


Road Network as a Graph

Nodes: Road segments

Edges: Connectivity between road segments

Prediction: Time of Arrival (ETA)



Traffic Prediction via GNN

Predicting Time of Arrival with GNNs (2020)

Used in Google Maps, 2020

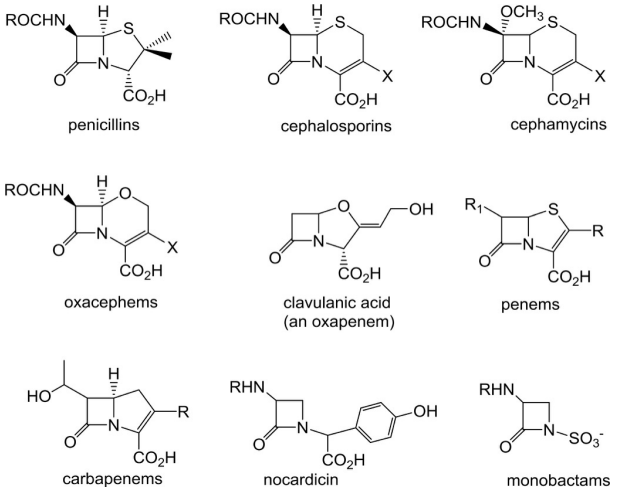
THE MODEL ARCHITECTURE FOR DETERMINING OPTIMAL ROUTES AND THEIR TRAVEL TIME.

Source: DeepMind, Sep 2020

Example (2): Drug Discovery

Antibiotics are small molecular graphs

- **Nodes:** Atoms
- **Edges:** Chemical bonds



Konaklieva, Monika I. "Molecular targets of β-lactam-based antimicrobials: beyond the usual suspects." Antibiotics 3.2 (2014): 128-142.

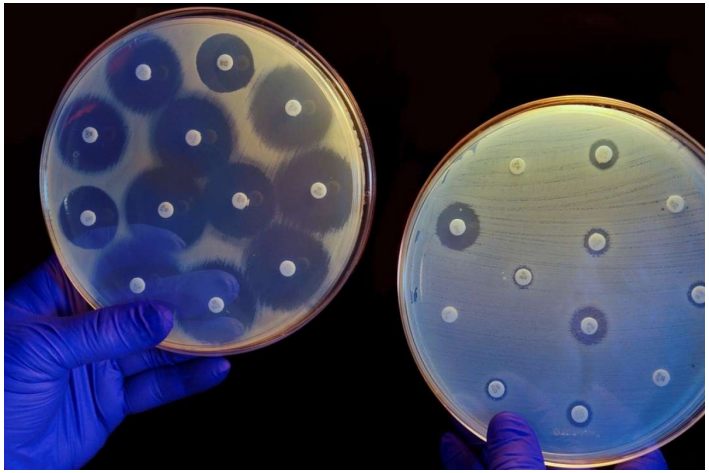


Image credit: CNN

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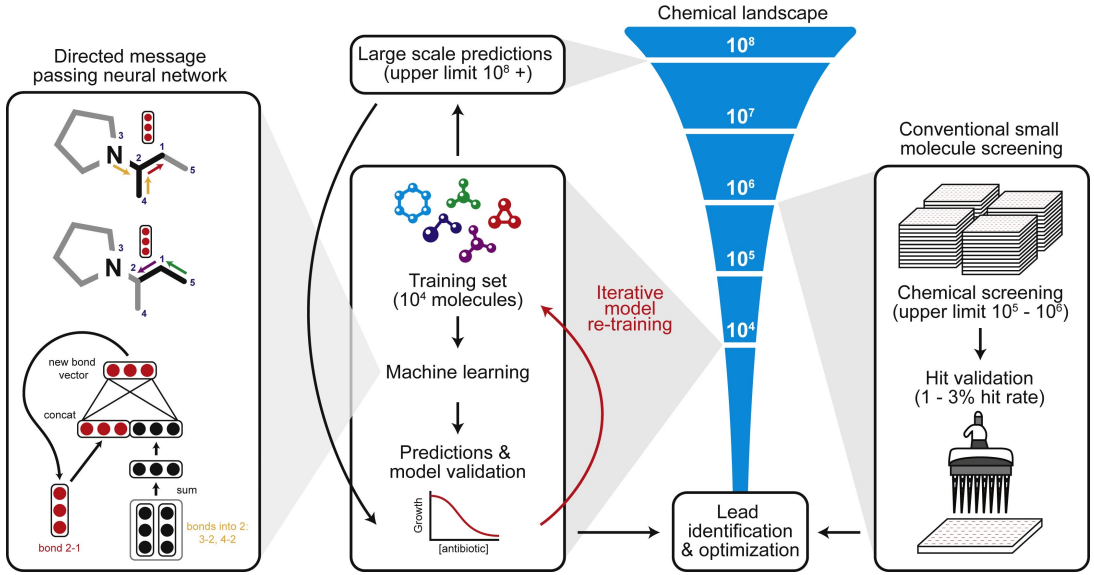
Deep Learning for Antibiotic Discovery

A Graph Neural Network **graph classification model**
Predict promising molecules from a pool of candidates



Stokes, Jonathan M., et al. "A deep learning approach to antibiotic discovery." Cell 180.4 (2020): 688-702.

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Therapeutics Data Commons

Artificial intelligence foundation for therapeutic scien

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Developing and Maintaining Therapeutics Data Commons

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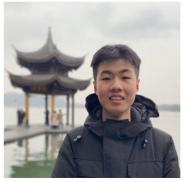

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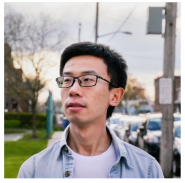

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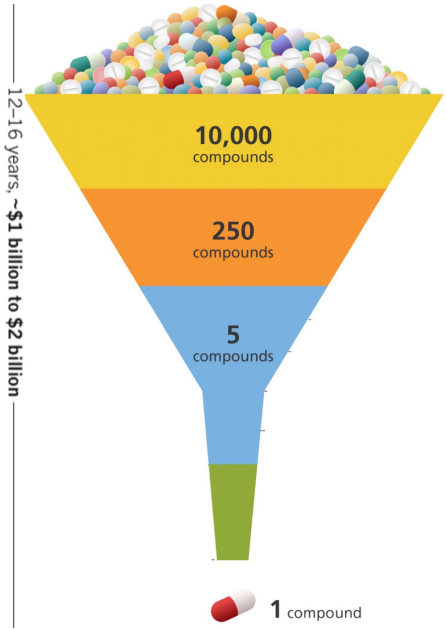
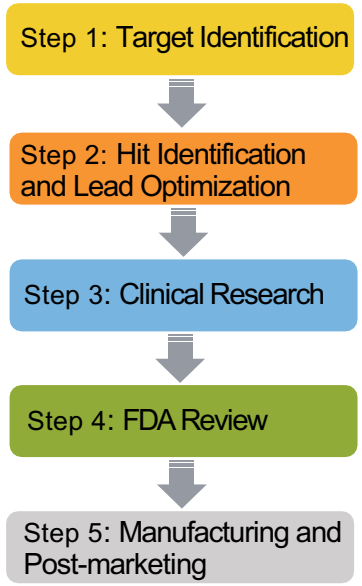
Challenges in Drug Discovery & Development

High cost, long time

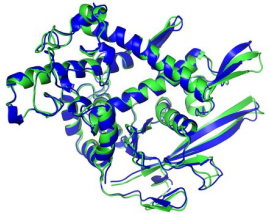
Various and emerging diseases

Abundant molecular modality

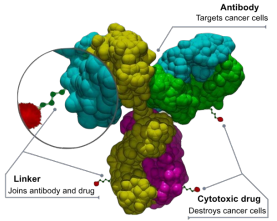




Small molecules

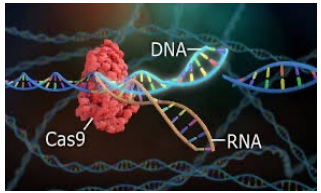


Proteins

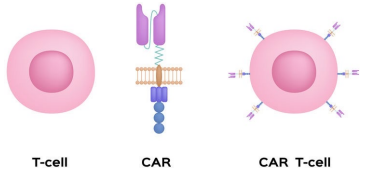


Anti- Nano-bodies

Vaccines



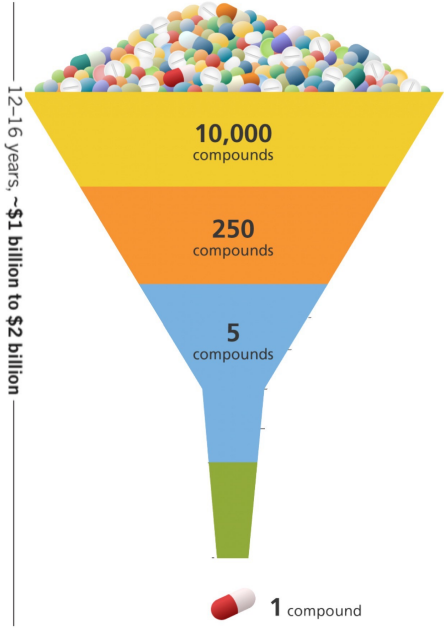
Gene-editing



CAR-T

Challenges of AI in Drug Discovery & Development

- Formulate various tasks as machine-learning-solvable tasks.
- Identify, retrieve, and process datasets of many different types scattered around.
- Assess algorithmic advances to align with real-world and clinical deployment.



Step 1: Target Identification → Gene-disease/drug interaction prediction; PPI prediction …

Step 2: Hit Identification and Lead Optimization → Molecular optimization; Protein structure prediction; Molecule property prediction …

Step 3: Clinical Research → Clinical trial recruitment; Drug repurposing …

Step 4: FDA Review → Automating document generation, review, and processing …

Step 5: Manufacturing and Post-marketing → Synthesis planning and prediction; Polypharmacy/food-drug prediction…

- Supervised learning
- Semi/self-supervised learning
- Generative modeling
- Optimization
- Reinforcement learning
- Interaction prediction
- Causality inferring
- …

Challenges of AI in Drug Discovery & Development

- Formulate various tasks as machine-learning-solvable tasks.
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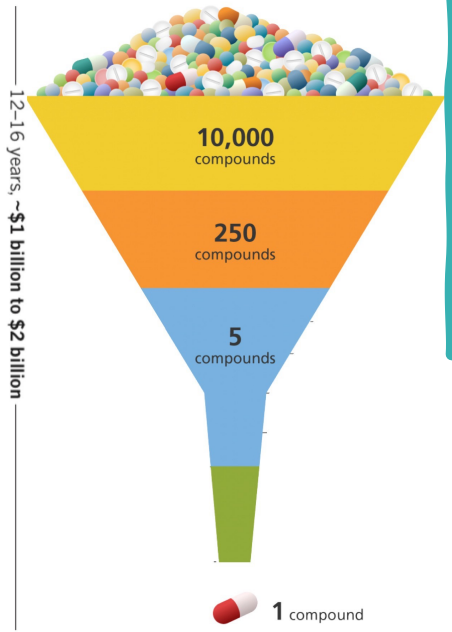

How to Advance Data-driven Therapeutics Design?

Step 3: Clinical Research

Clinical trial recruitment; Drug

Step 4: FDA Review

Automating document generation, review, and processing …

Step 5: Manufacturing and Post-marketing

Synthesis planning and prediction; Polypharmacy/food-drug prediction…

- Supervised learning
- Semi/self-supervised learning
- Generative modeling
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- Causality inferring
- …

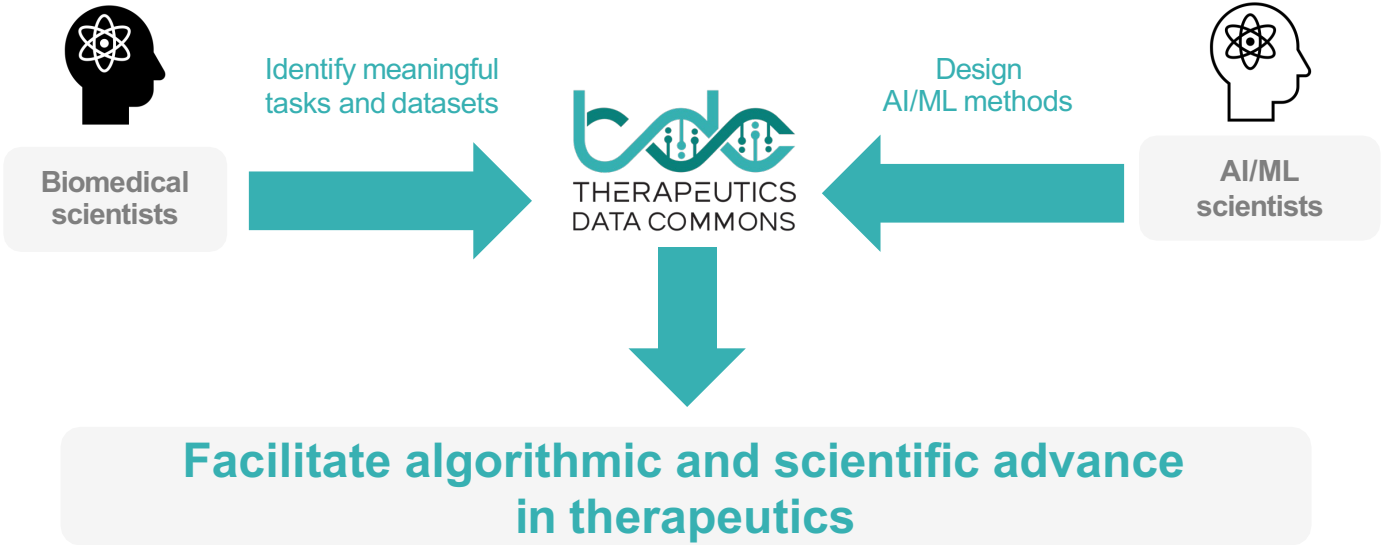
Therapeutics Data Commons

THERAPEUTICS DATA COMMONS

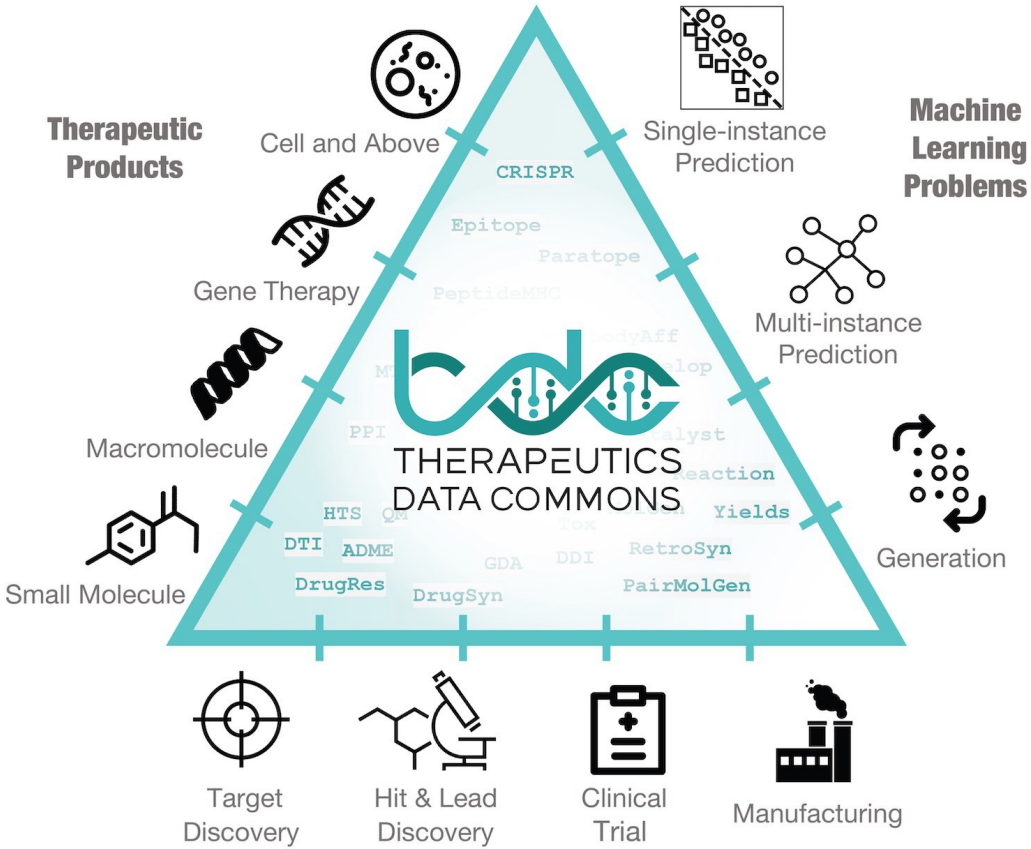
Website: https://tdcommons.ai (or QR code)

Paper: https://www.nature.com/articles/s41589-022-01131-2

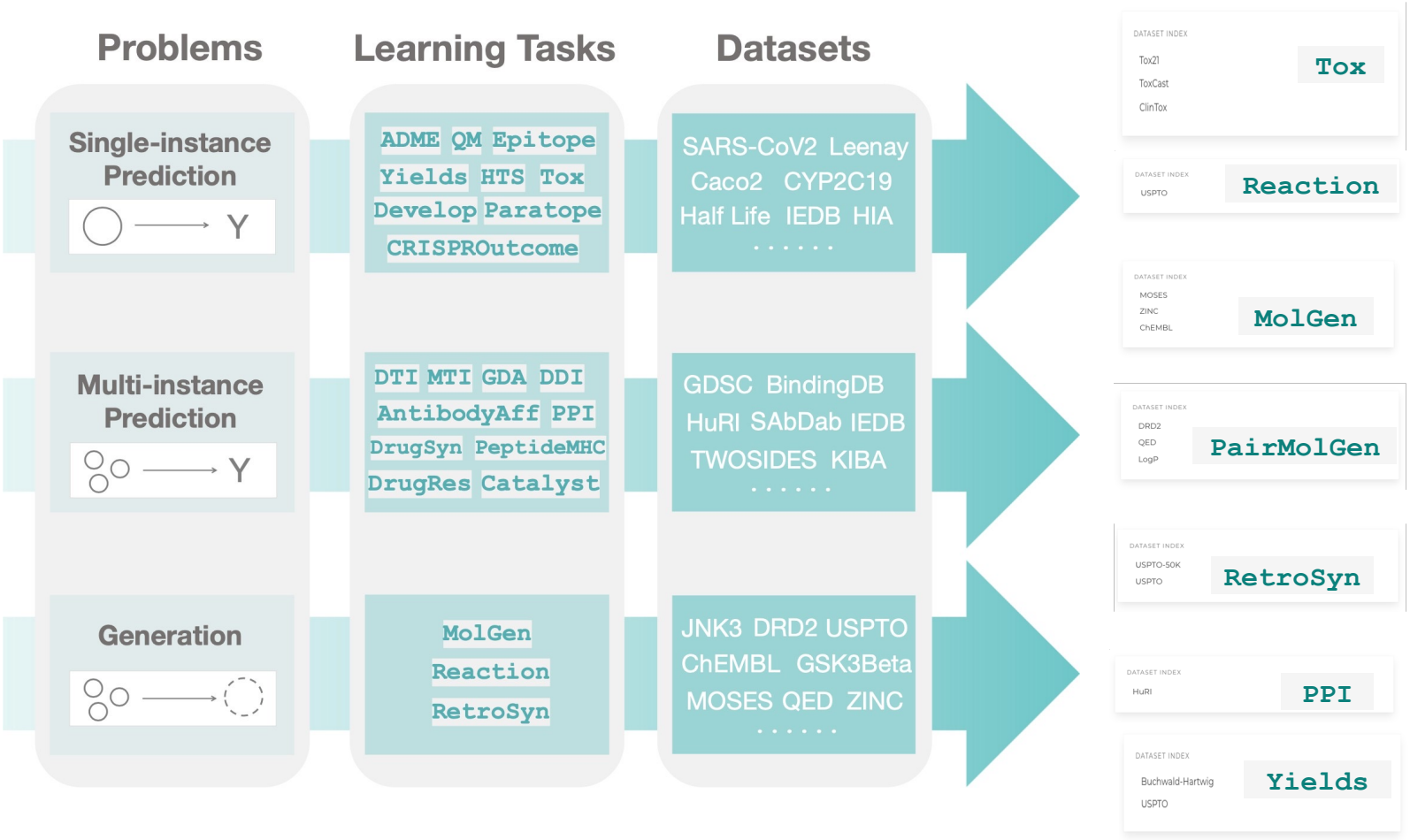
Github: https://github.com/mims-harvard/TDC



Wide Range of Therapeutic Modalities and Pipeline



Three-Tier Design



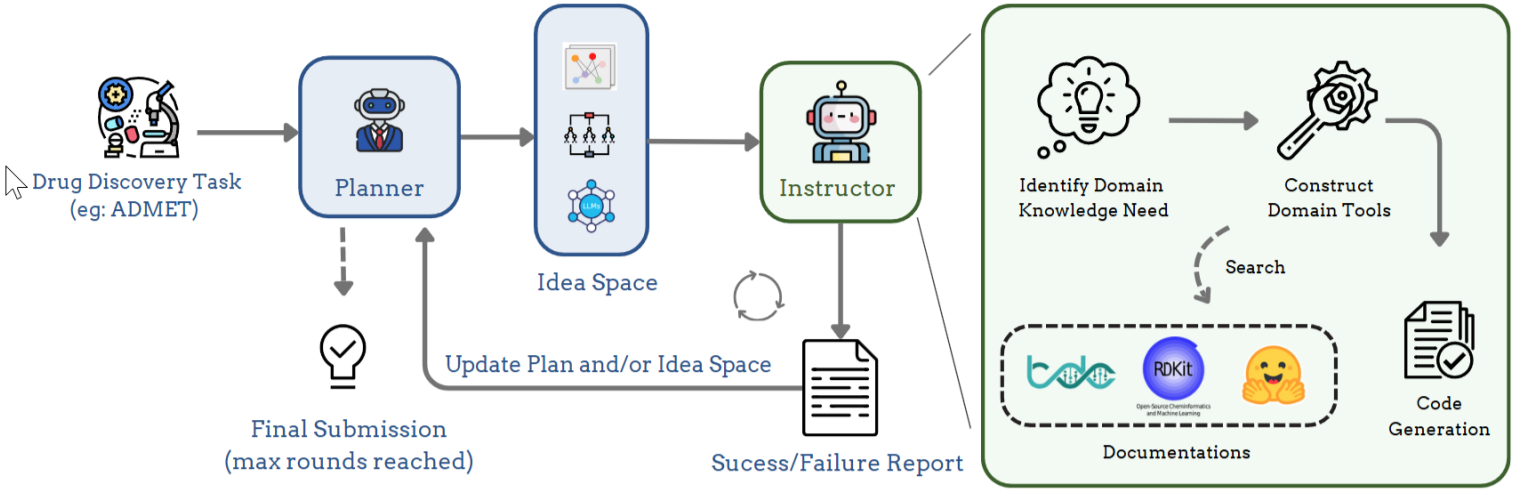


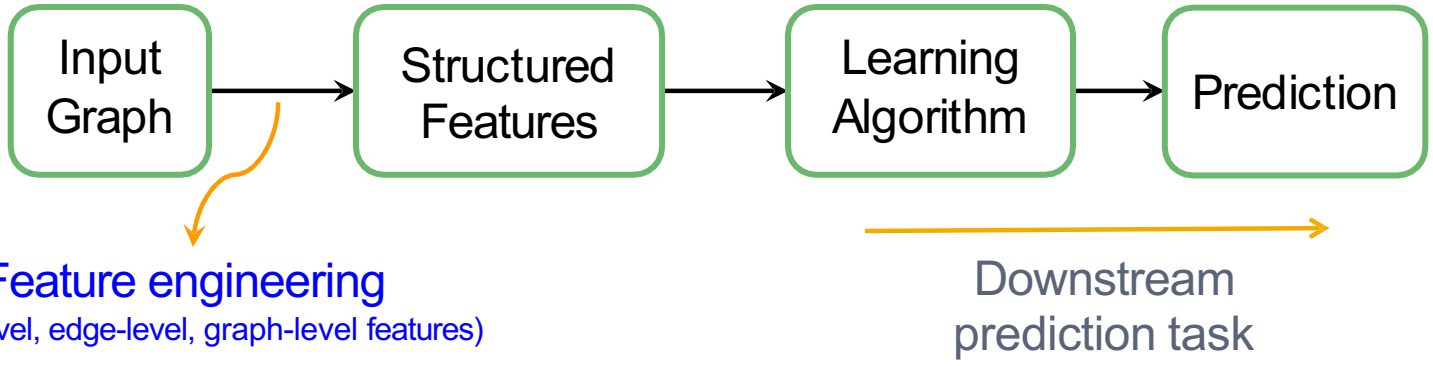
Figure 1: Framework overview of DrugAgent. Given an AI-based drug discovery task described in natural language (i.e., user's input, e.g., design an AI model to predict Absorption (one of the ADMET properties) using the PAMPA dataset (Siramshetty et al., 2021), the Planner first produces a couple of potential ideas (e.g., GCN (graph convolutional network) (Kipf and Welling, 2016), random forest, pretrained model (such as ChemBERTa (Chithrananda et al., 2020))). Then, the Planner pass an idea to the Instructor, who transfers the idea into code by integrating domain knowledge (e.g., *molecular fingerprinting* and *molecular graph construction*) through doc search. The Instructor reports the results of the investigation back to the Planner, which uses this feedback to update plan and begin the next round of exploration. Finally, the Planner collects all results and reports the best-performing idea (e.g., random forest achieves the best performance in predicting *absorption*).

Graph Neural Networks (GNN)

Details

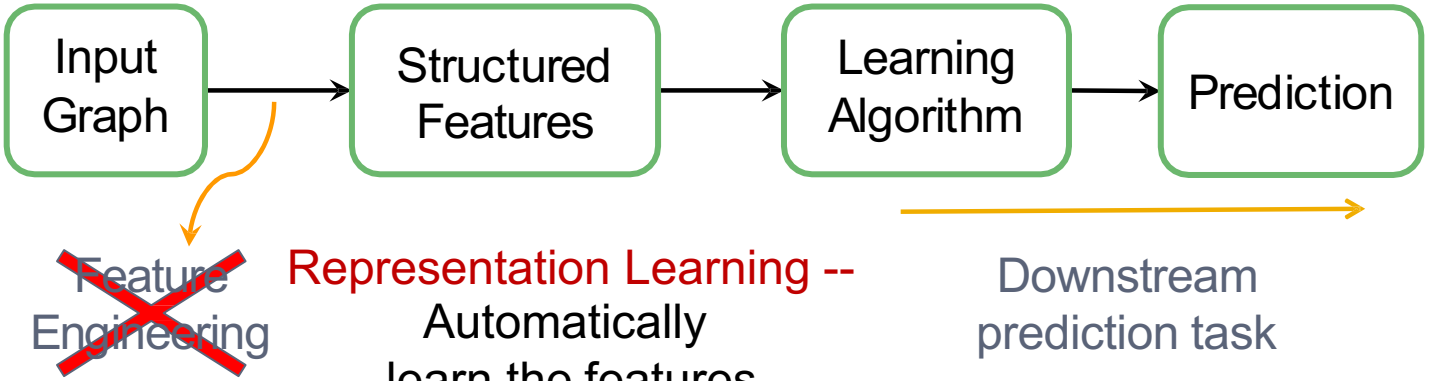
Recap: Traditional ML for Graphs

Given an input graph, extract node, link and graph-level features, then learn a model (SVM, neural network, etc.) that maps features to labels.

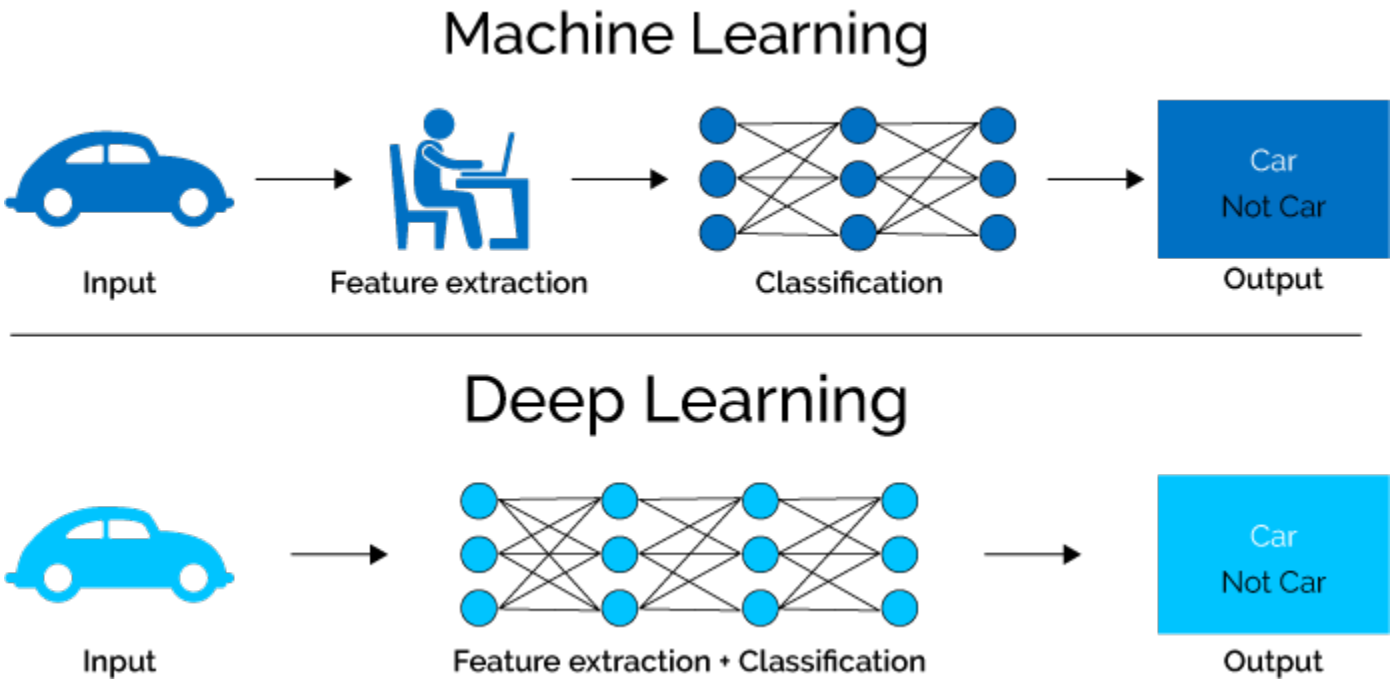


Graph Representation Learning

Graph Representation Learning alleviates the need to do feature engineering every single time.

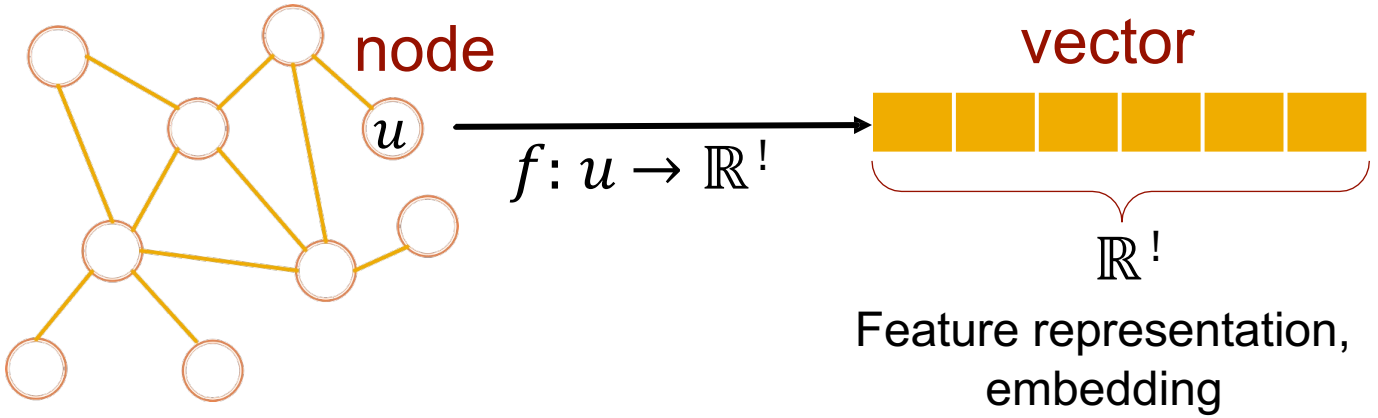


Recap: Classical ML vs. DL (Better Representations)



Graph Representation Learning

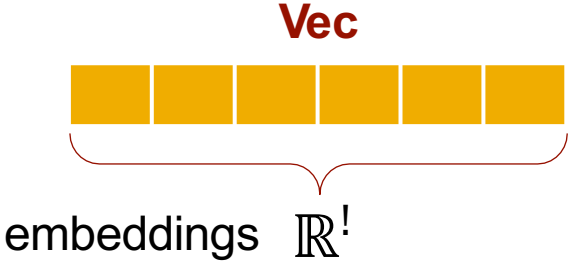
Goal: Efficient task-independent feature learning for machine learning with graphs!

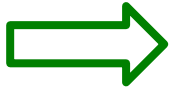


Why Embedding?

Task: Map nodes into an embedding space

- Similarity of embeddings between nodes indicates their similarity in the network. For example:
  - Both nodes are close to each other (connected by an edge)
- Encode network information
- Potentially used for many downstream predictions





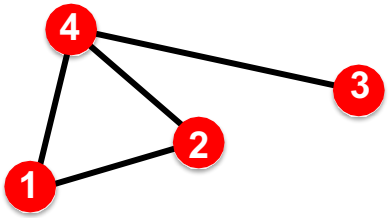
Tasks

- Node classification
- Link prediction
- Graph classification
- Anomalous node detection
- Clustering
- ….

Setup

Assume we have an (undirected) graph G:

- V is the vertex set.
- $\mathbf{A}$ is the adjacency matrix (assume binary).
- **For simplicity: No node features or extra information is used**

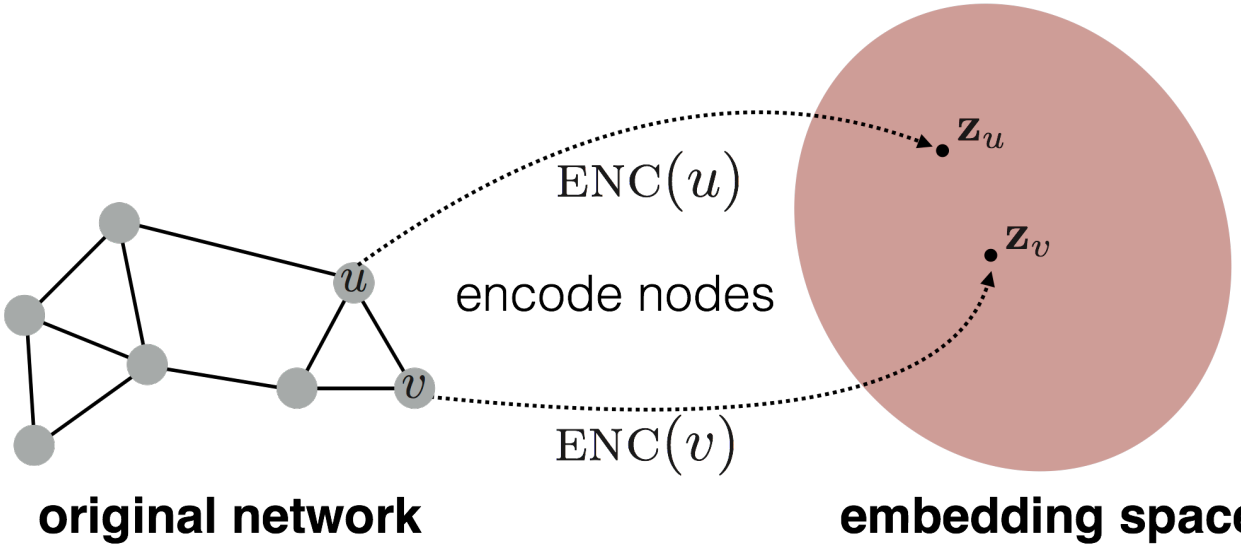


V: {1, 2, 3, 4}

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Embedding Nodes

Goal is to encode nodes so that similarity in the embedding space (e.g., dot product) approximates similarity in the graph



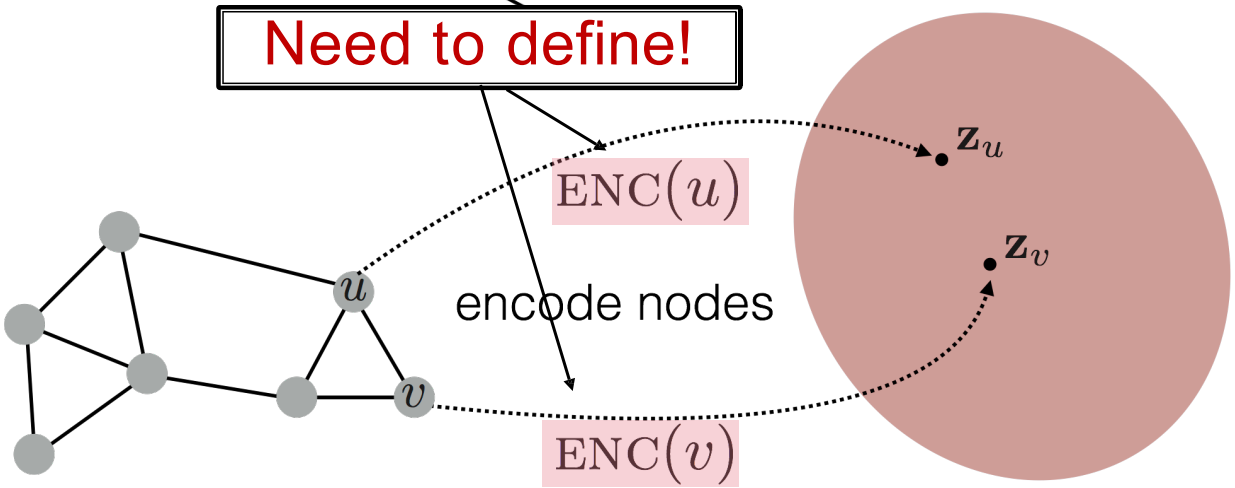
Embedding Nodes

Goal: $\text{similarity}(u, v) \approx \mathbf{z}_v^{\mathrm{T}} \mathbf{z}_u$

similarity(u, v): in the original network

$\mathbf{z}_v^{\mathrm{T}} \mathbf{z}_u$: Similarity of the embedding

Need to define!



Learning Node Embeddings

1. **Encoder** maps from nodes to embeddings
2. **Define a node similarity function** (i.e., a measure of similarity in the original network)
3. **Decoder DEC** maps from embeddings to the similarity score
4. **Optimize the parameters of the encoder so that:**

$\mathrm{DEC}(\mathbf{z}_v^{\mathrm{T}}\mathbf{z}_u)$

$$\underbrace{\text{similarity}(u, v)}_{\text{in the original network}} \approx \underbrace{\mathbf{z}_v^{\mathrm{T}}\mathbf{z}_u}_{\text{Similarity of the embedding}}$$

Two Key Components

Encoder: maps each node to a low-dimensional vector

$$\text{ENC}(v) = \mathbf{z}_!$$

d-dimensional embedding

node in the input graph

Similarity function: specifies how the relationships in vector space map to the relationships in the original network

$$\text{similarity}(u, v) \approx \mathbf{z}_v^{\mathrm{T}} \mathbf{z}_u$$

Decoder

Similarity of u and v in the original network

dot product between node embeddings

"Shallow" Encoding

Simplest encoding approach: Encoder is just an embedding-lookup

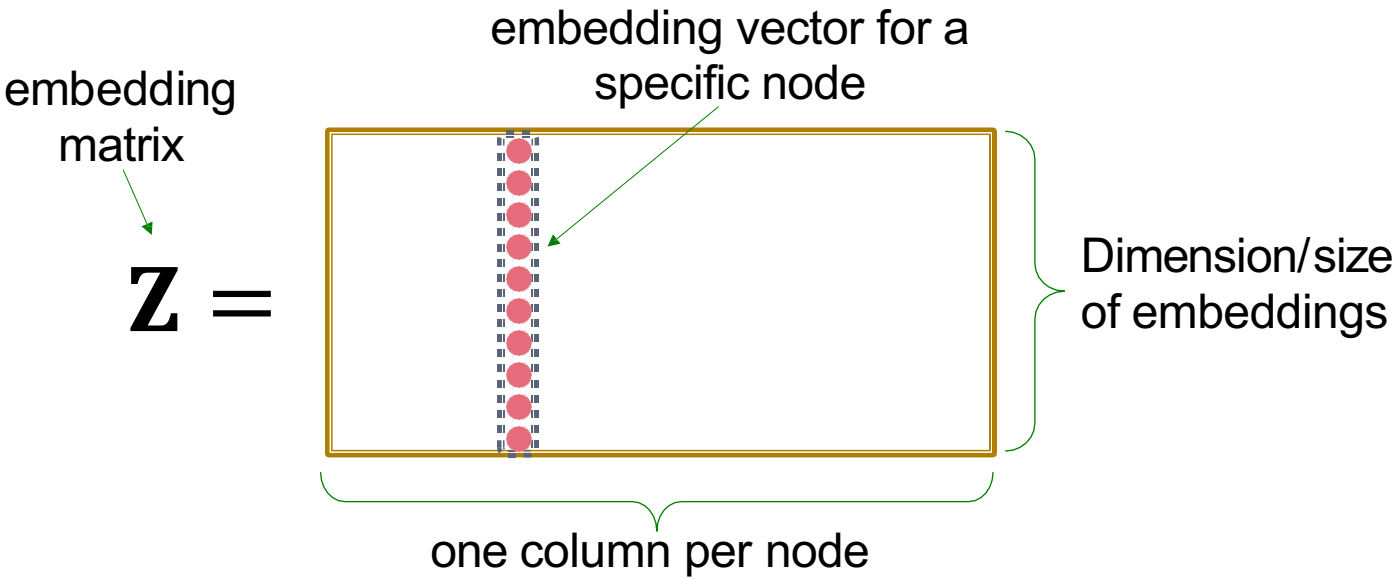
$$\text{ENC}(v) = \mathbf{z}_v = \mathbf{Z} \cdot v$$

$\mathbf{Z} \in \mathbb{R}^{d \times |\mathcal{V}|}$ matrix, each column is a node embedding [what we learn / optimize]

$v \in \mathbb{I}^{|\mathcal{V}|}$ indicator vector, all zeroes except a one in column indicating node *v*

"Shallow" Encoding

Simplest encoding approach: encoder is just an embedding-lookup



"Shallow" Encoding

Simplest encoding approach: **Encoder is just an embedding-lookup**

Each node is assigned a unique embedding vector
(i.e., we directly optimize the embedding of each node)

Many methods: DeepWalk, node2vec

Framework Summary

Encoder + Decoder Framework

- Shallow encoder: Embedding lookup
- Parameters to optimize: $\mathbf{Z}$ which contains node embeddings $\mathbf{z}_u$ for all nodes $u \in V$
- We will cover deep encoders in the GNNs in 20 mins

- **Decoder:** based on node similarity.
- **Objective:** maximize $\mathbf{z}_v^{\mathrm{T}} \mathbf{z}_u$ for node pairs (u, v) that are **similar**

How to Define Node Similarity?

Key choice of methods is **how they define node similarity.**

Should two nodes have a similar embedding if they…

- are linked?
- share neighbors?
- have similar “structural roles”?

Note on Node Embeddings

This is **unsupervised/self-supervised** way of learning node embeddings.

- We are **not** utilizing node labels
- We are **not** utilizing node features
- The goal is to directly estimate a set of coordinates (i.e., the embedding) of a node so that some aspect of the network structure is preserved.

These embeddings are **task independent:**

- They are not trained for a specific task but can be used for any task.

Today: Deep Graph Encoders

Now: discuss deep learnig methods based on **graph neural networks (GNNs):**

$$\text{ENC}(v) = \text{multiple layers of non-linear transformations based on graph structure}$$

Note: All these deep encoders can be **combined with node similarity functions** defined earlier.

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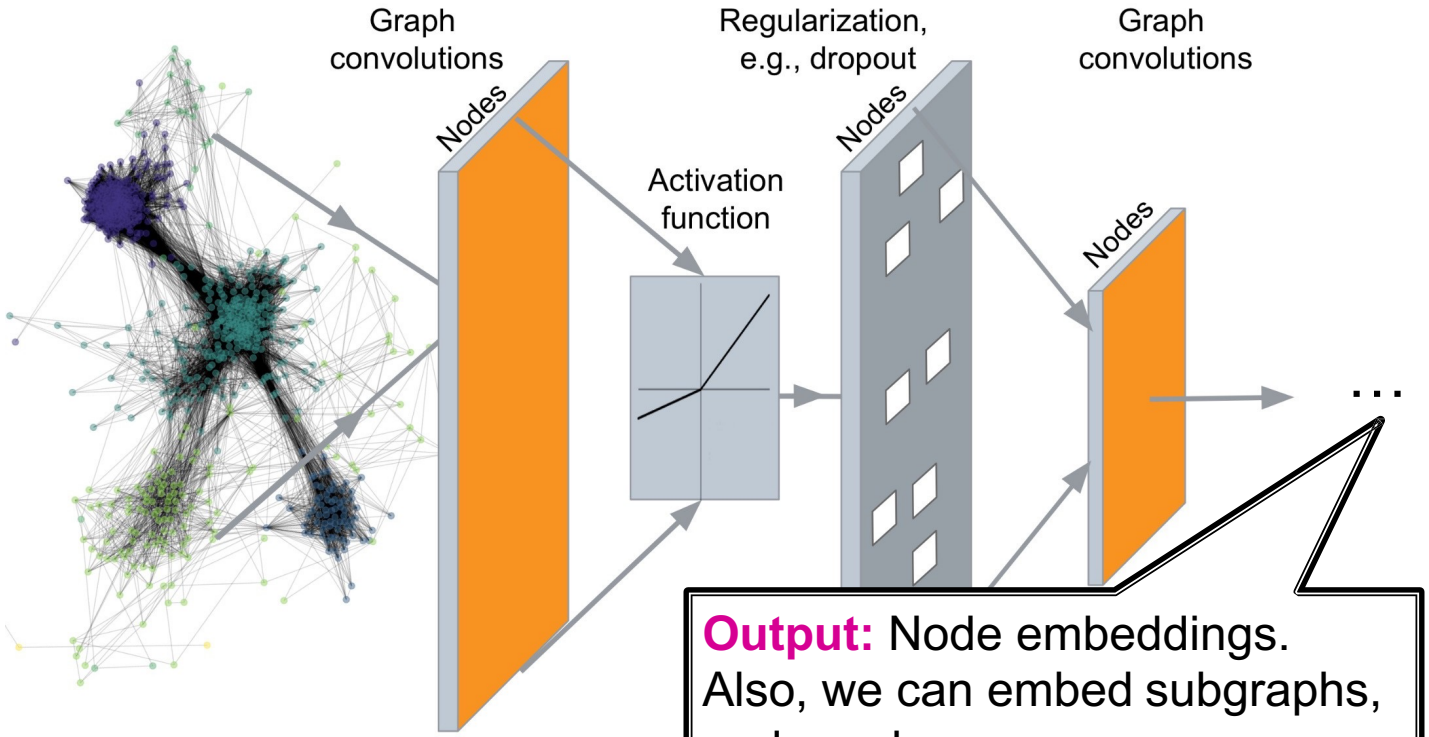
Deep Graph Encoders



Output: Node embeddings. Also, we can embed subgraphs, and graphs

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Tasks on Networks

Tasks we will be able to solve:

Node classification

- Predict the type of a given node

Link prediction

- Predict whether two nodes are linked

Community detection

- Identify densely linked clusters of nodes

Network similarity

- How similar are two (sub)networks

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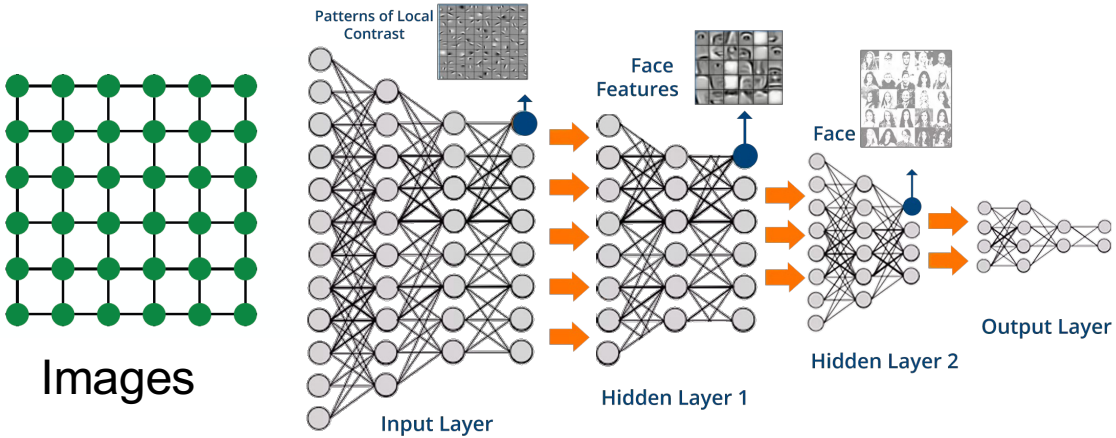
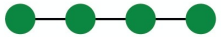
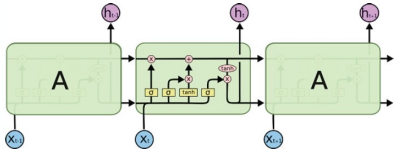
Modern ML Toolbox





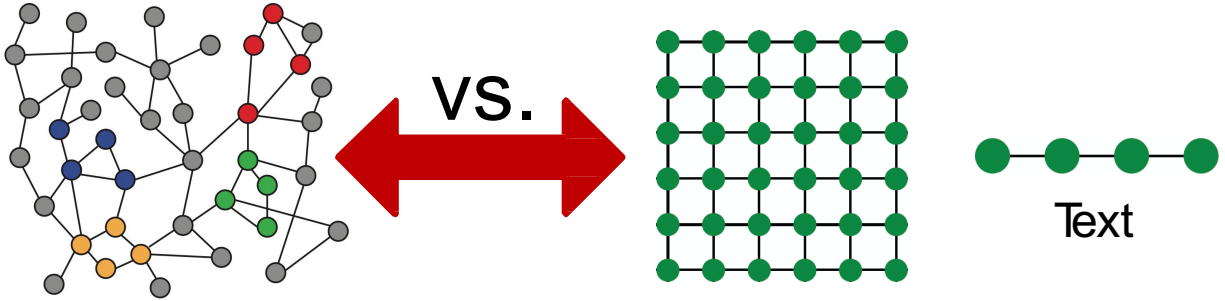


Modern deep learning toolbox is designed for simple sequences & grids

Why is it Hard?

But networks are far more complex!

- Arbitrary size and complex topological structure (i.e., no spatial locality like grids)



- No fixed node ordering or reference point
- Often dynamic and have multimodal features

Summary: Basics of Deep Learning

Loss function:

$$\min_{\Theta} \mathcal{L}(\boldsymbol{y}, f_{\Theta}(\boldsymbol{x}))$$

f can be a simple linear layer, an MLP, or other neural networks (e.g., a GNN later)

Sample a minibatch of input $\boldsymbol{x}$

Forward propagation: Compute $\mathcal{L}$ given $\boldsymbol{x}$

Back-propagation: Obtain gradient $\nabla_{\Theta}\mathcal{L}$ using a chain rule.

Use **stochastic gradient descent (SGD)** to optimize $\mathcal{L}$ for Θ over many iterations.

Content

Local network neighborhoods:

- Describe aggregation strategies
- Define computation graphs

Stacking multiple layers:

- Describe the model, parameters, training
- How to fit the model?
- Simple example for unsupervised and supervised training

Setup

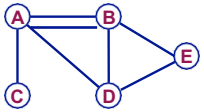
Assume we have a graph G:

- V is the **vertex set**
- $\boldsymbol{A}$ is the **adjacency matrix** (assume binary)
- $\boldsymbol{X} \in \mathbb{R}^{|V| \times m}$ is a matrix of **node features**
- v: a node in V; $N(v)$: the set of neighbors of v.
- **Node features:**
  - Social networks: User profile, User image
  - Biological networks: Gene expression profiles, gene functional information
  - When there is no node feature in the graph dataset:
    - Indicator vectors (one-hot encoding of a node)
    - Vector of constant 1: [1, 1, …, 1]

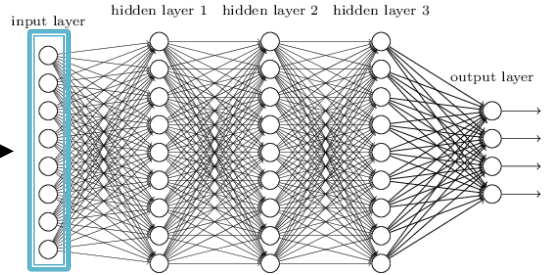
A Naïve Approach

Join adjacency matrix and features

Feed them into a deep neural net:



	A	B	C	D	E	Feat	
A	0	1	1	1	0	1	0
B	1	0	0	1	1	0	0
C	1	0	0	1	0	0	1
D	1	1	1	0	1	1	1
E	0	1	0	1	0	1	0



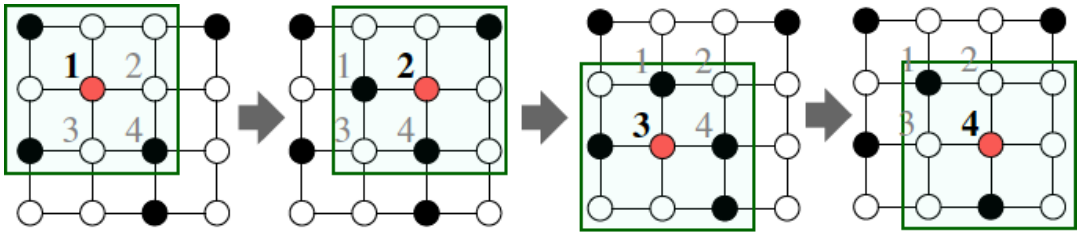
?

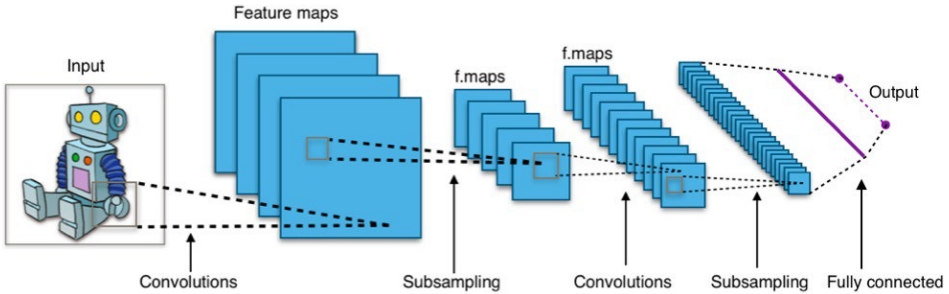
Issues with this idea:

- $O(|V|)$ parameters
- Not applicable to graphs of different sizes
- Sensitive to node ordering

Idea: Convolutional Networks

CNN on an image:



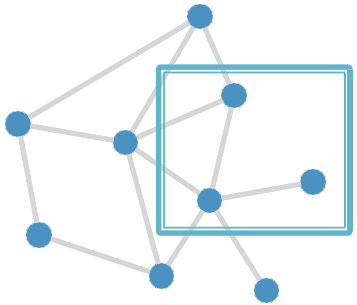


Goal is to generalize convolutions beyond simple lattices

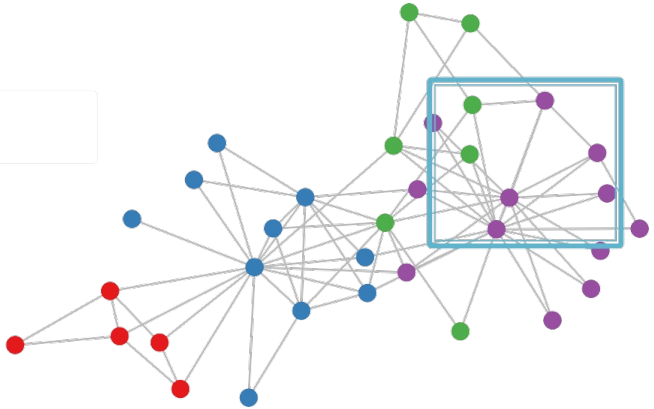
Leverage node features/attributes (e.g., text, images)

Real-World Graphs

But our graphs look like this:



or this:



- There is no fixed notion of locality or sliding window on the graph
- Node numbering is arbitrary

Permutation Invariance

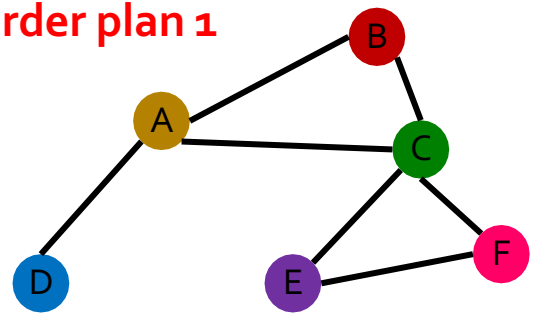
Graph does not have a canonical order of the nodes!

We can have many different order plans.

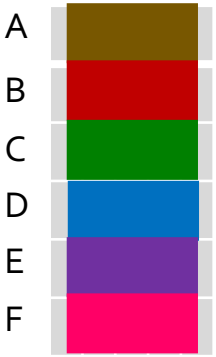
Permutation Invariance

Graph does not have a canonical order of the nodes!

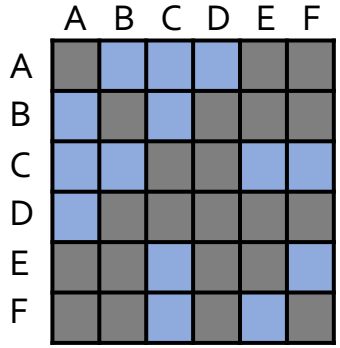
Order plan 1



Node features X_1



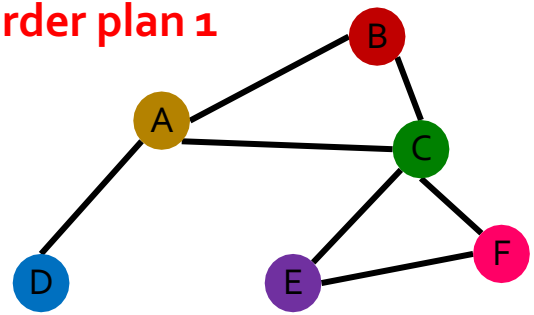
Adjacency matrix A_1



Permutation Invariance

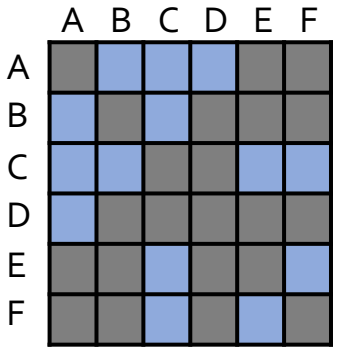
Graph does not have a canonical order of the nodes!

Order plan 1

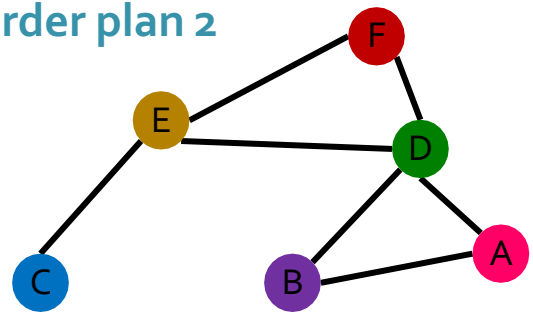


Node features X_1

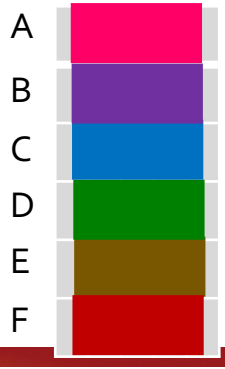
Adjacency matrix A_1



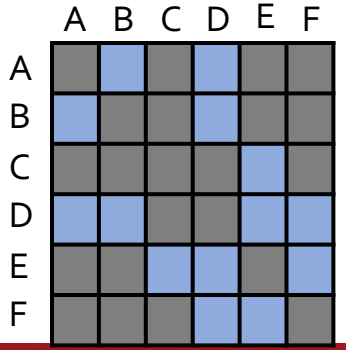
Order plan 2



Node features X_2



Adjacency matrix A_2



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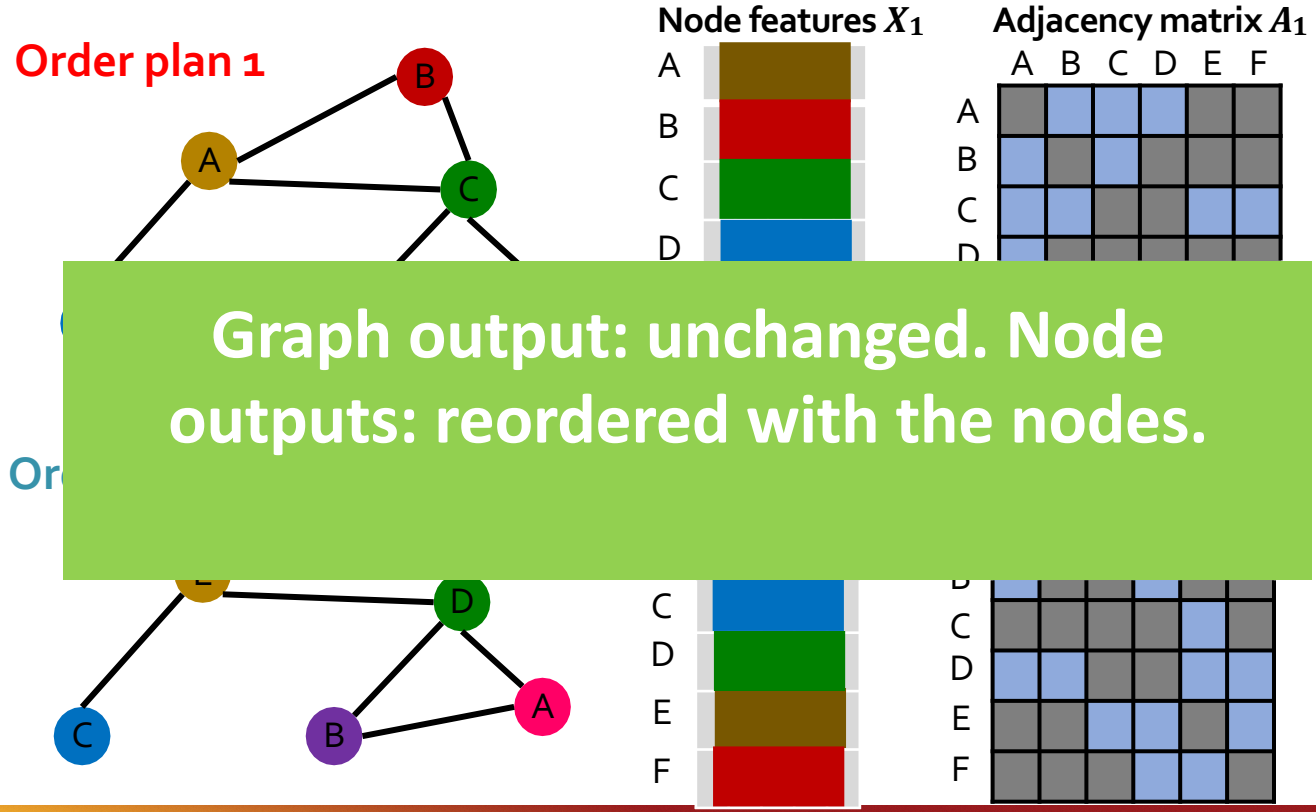
Permutation Invariance

Graph does not have a canonical order of the nodes!



Permutation Invariance

What does it mean by "graph representation is same for two order plans"?

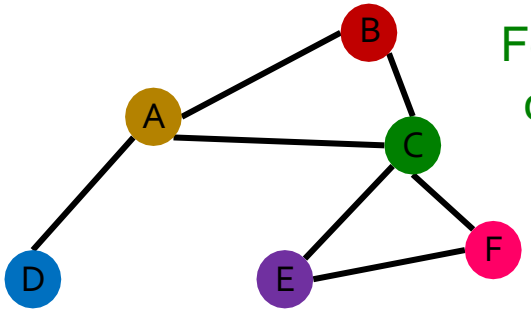
In other words, f maps a graph to a d-dim embedding

Consider we learn a function f that maps a graph $G = (\boldsymbol{A}, \boldsymbol{X})$ to a vector $\mathbb{R}^{\mathrm{d}}$ then

$$f(\boldsymbol{A}_1, \boldsymbol{X}_1) = f(\boldsymbol{A}_2, \boldsymbol{X}_2)$$

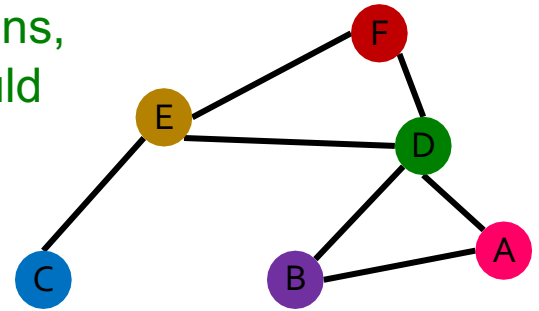
$\boldsymbol{A}$ is the adjacency matrix
$\boldsymbol{X}$ is the node feature matrix

Order plan 1: $\boldsymbol{A}_1, \boldsymbol{X}_1$



For two order plans, output of f should be the same!

Order plan 2: $\boldsymbol{A}_2, \boldsymbol{X}_2$



Permutation Invariance

What does it mean by "graph representation is same for two order plans"?

- Consider we learn a function f that maps a graph $G = (\boldsymbol{A}, \boldsymbol{X})$ to a vector $\mathbb{R}^d$.

  $\boldsymbol{A}$ is the adjacency matrix
  $\boldsymbol{X}$ is the node feature matrix

- Then, if $f(\boldsymbol{A}_i, \boldsymbol{X}_i) = f(\boldsymbol{A}_j, \boldsymbol{X}_j)$ for any order plan i and j, we formally say f is a **permutation invariant function**.

  For a graph with $|V|$ nodes, there are $|V|!$ different order plans.

  m… each node has a m-dim feature vector associated with it.

- **Definition:** For any graph function $f: \mathbb{R}^{|V| \times m} \times \mathbb{R}^{|V| \times |V|} \rightarrow \mathbb{R}^d$, f is **permutation-invariant** if $f(A, X) = f(PAP^T, PX)$ for any permutation P.

  d… output embedding dimensionality of embedding the graph $G = (A, X)$

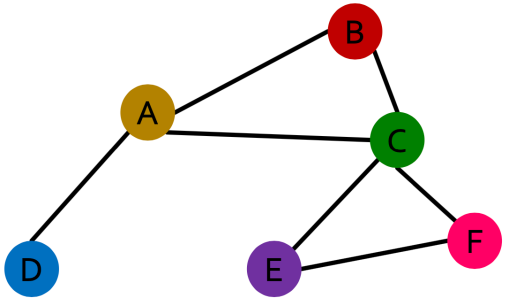
  Permutation P: a shuffle of the node order
  Example: (A,B,C)->(B,C,A)

Permutation Equivariance

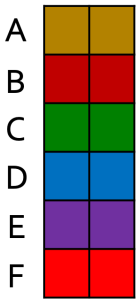
For node representation: We learn a function f that maps nodes of G to a matrix $\mathbb{R}^{|V|\times d}$.

In other words, each node in V is mapped to a d-dim embedding.

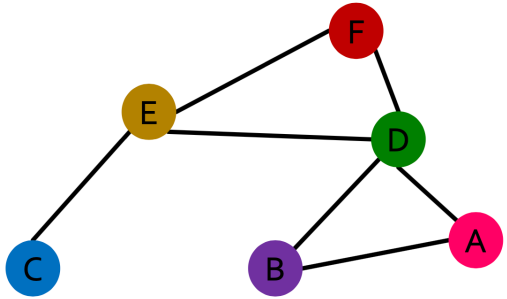
Order plan 1: A_1, X_1



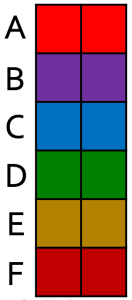
$$f(\boldsymbol{A_1}, \boldsymbol{X_1}) =$$



Order plan 2: A_2, X_2



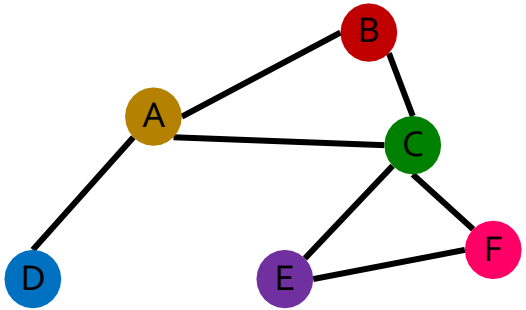
$$f(\boldsymbol{A_2}, \boldsymbol{X_2}) =$$



Permutation Equivariance

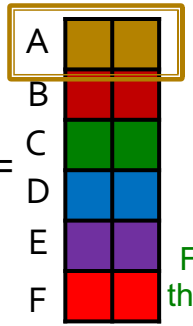
For node representation: We learn a function f that maps nodes of G to a matrix $\mathbb{R}^{|v| \times d}$.

Order plan 1: A_1, X_1

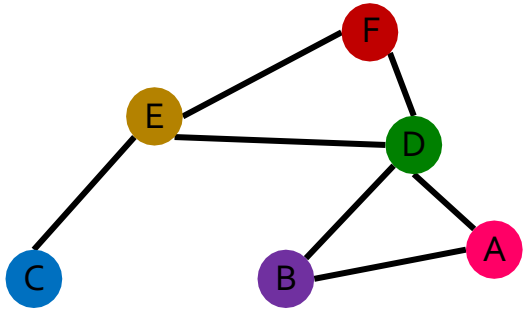


Representation vector of the brown node A

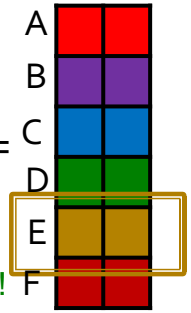
$f(A_1, X_1) =$



Order plan 2: A_2, X_2



$f(A_2, X_2) =$



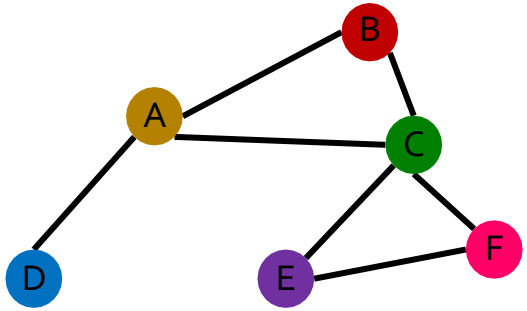
Representation vector of the brown node E

For two order plans, the vector of node at the same position in the graph is the same!

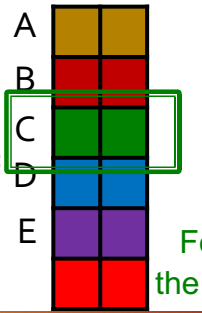
Permutation Equivariance

For node representation: We learn a function f that maps nodes of G to a matrix $\mathbb{R}^{|V| \times d}$.

Order plan 1: A_1, X_1

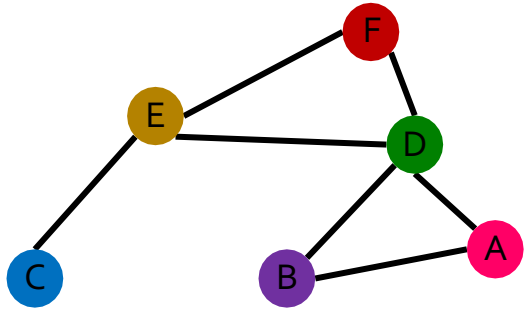


$f(A_1, X_1) =$

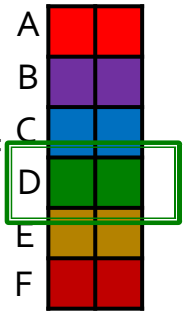


Representation vector of the green node C

Order plan 2: A_2, X_2



$f(A_2, X_2) =$



Representation vector of the green node D

For two order plans, the vector of node at the same position in the graph is the same!

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Permutation Equivariance

For node representation:

- Consider we learn a function f that maps a graph $G = (\boldsymbol{A}, \boldsymbol{X})$ to a matrix $\mathbb{R}^{|V| \times d}$
- If the output vector of a node at the same position in the graph remains unchanged for any order plan, we say f is **permutation equivariant**.

m… each node has a m-dim feature vector associated with it.

- **Definition:** For any node function $f: \mathbb{R}^{|V| \times m} \times \mathbb{R}^{|V| \times |V|} \rightarrow \mathbb{R}^{|V| \times d}$, f is **permutation-equivariant** if $Pf(A, X) = f(PAP^T, PX)$ for any permutation P.

f maps each node in V to a d-dim embedding.

Summary: Invariance and Equivariance

- **Permutation-invariant**

$$f(A, X) = f(PAP^T, PX)$$

Permute the input, the output stays the same. (map a graph to a vector)

graph level

- **Permutation-equivariant**

$$Pf(A, X) = f(PAP^T, PX)$$

Permute the input, output also permutes accordingly. (map a graph to a matrix)

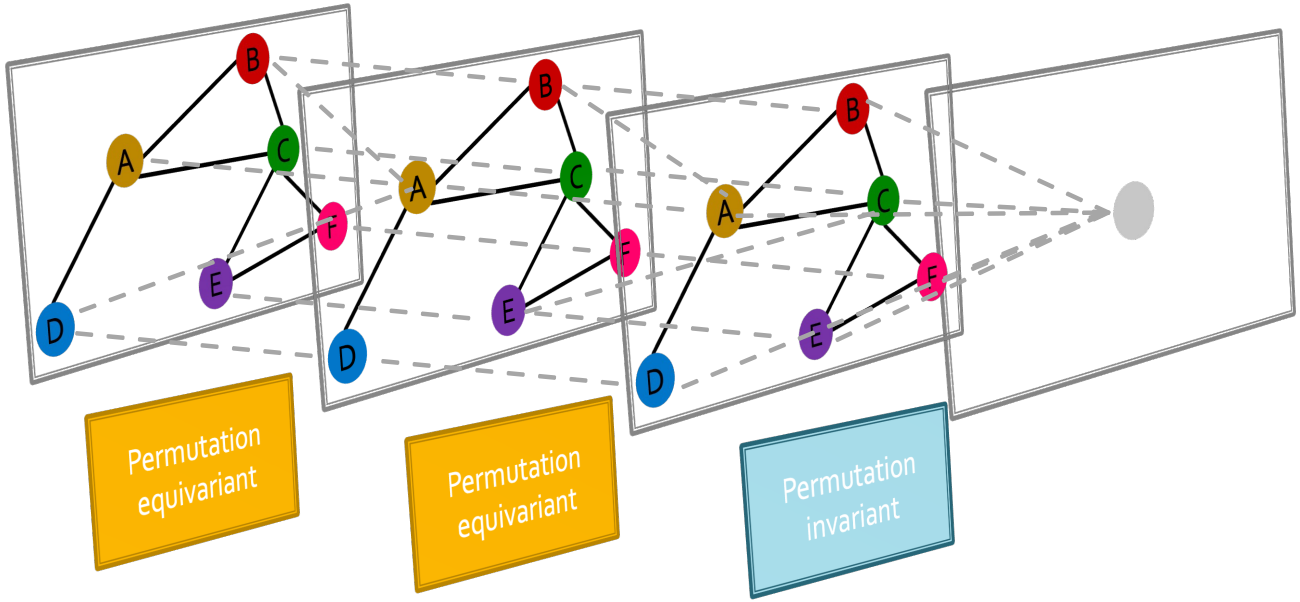
node level

- **Examples:**
  - $f(A, X) = 1^T X$: Permutation-**invariant**
    - Reason: $f(PAP^T, PX) = 1^T PX = 1^T X = f(A, X)$
  - $f(A, X) = X$: Permutation-**equivariant**
    - Reason: $f(PAP^T, PX) = PX = Pf(A, X)$
  - $f(A, X) = AX$: Permutation-**equivariant**
    - Reason: $f(PAP^T, PX) = PAP^T PX = PAX = Pf(A, X)$

[Bronstein, ICLR 2021 keynote]

Graph Neural Network Overview

Graph neural networks consist of multiple permutation equivariant / invariant functions.



Flattened Slots Can Break Symmetry

Graph-level predictor: $y = 1 \cdot \text{slot}_1 + 2 \cdot \text{slot}_2$

Order A, B
[1, 3] → y = 7

Order B, A
[3, 1] → y = 5

Different slot weights make the graph output depend on numbering.

Course schematic.

A Shared Node MLP Is Equivariant

Apply the same f to every node independently.

Order A, B
[f(1), f(3)]

Order B, A
[f(3), f(1)]

The rows reorder correctly, but no node reads its neighbors.

Shared node MLP + neighbor aggregation gives a graph-aware update.

Course schematic.

Graph Neural Network Overview

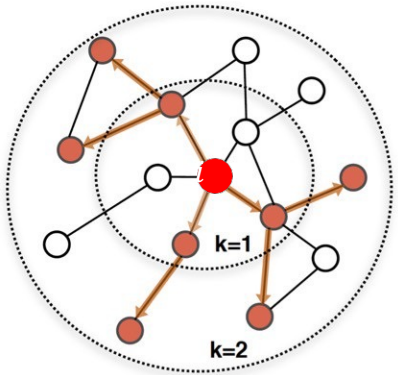
Shared node MLPs still ignore edges.

Next: Design graph neural networks that are permutation invariant / equivariant by passing and aggregating information from neighbors!

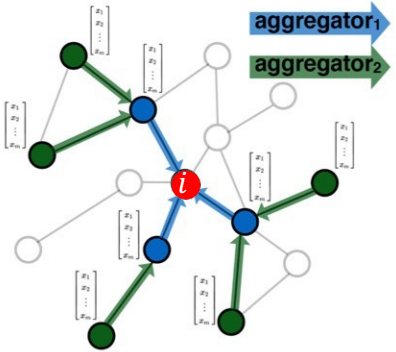
?

Graph Convolutional Networks

Idea: Node's neighborhood defines a computation graph



Determine node computation graph

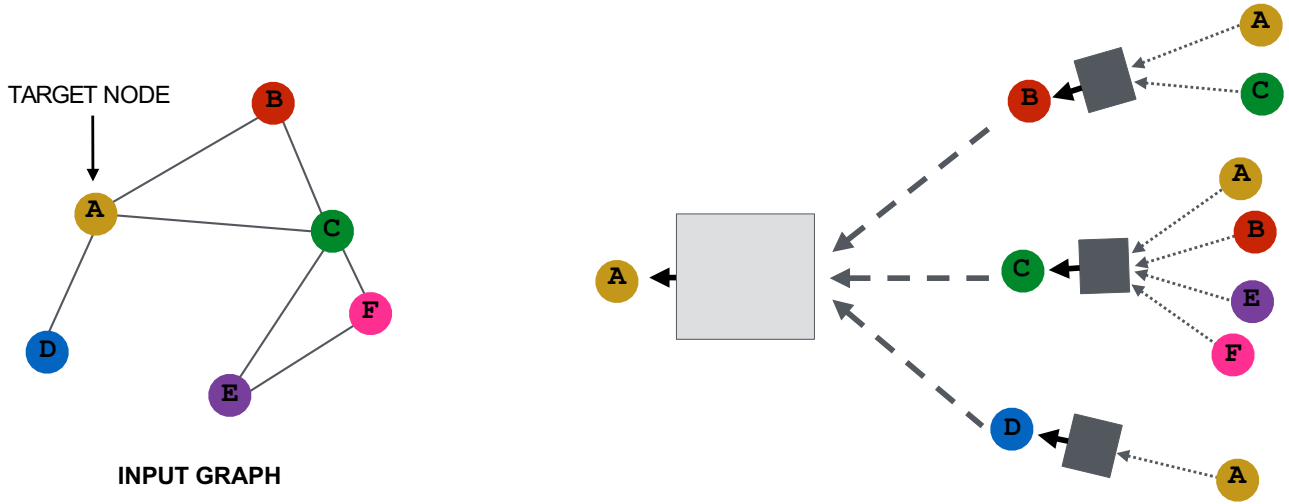


Propagate and transform information

Learn how to propagate information across the graph to compute node features

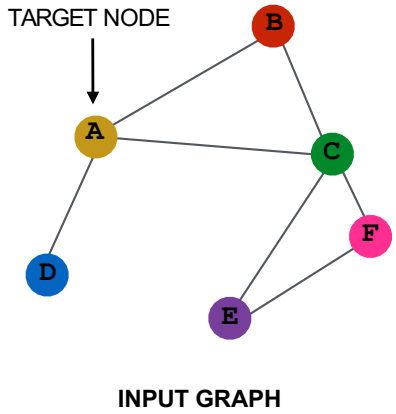
Idea: Aggregate Neighbors

Key idea: Generate node embeddings based on **local network neighborhoods**

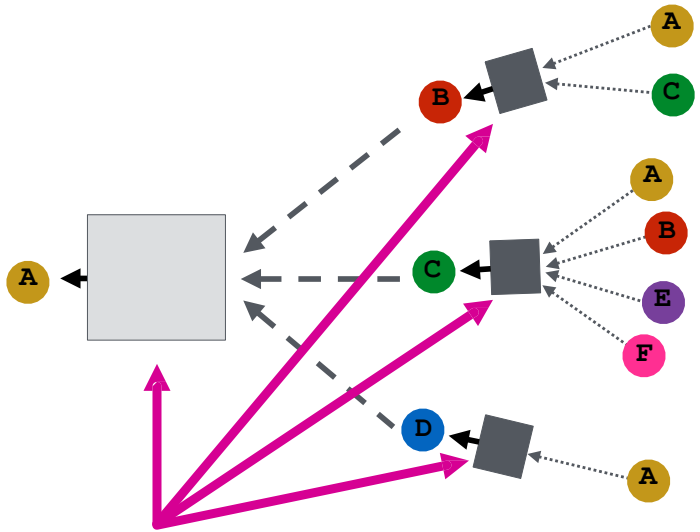


Idea: Aggregate Neighbors

Intuition: Nodes aggregate information from their neighbors using neural networks



INPUT GRAPH

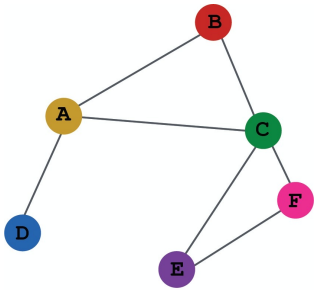


Neural networks

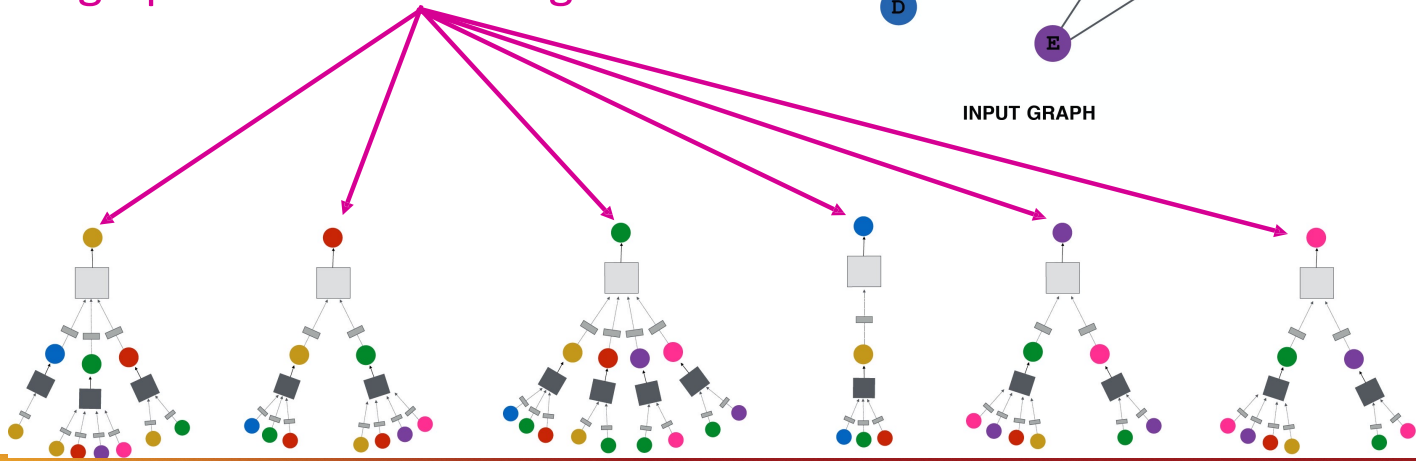
Idea: Aggregate Neighbors

Intuition: Network neighborhood defines a computation graph

Every node defines a computation graph based on its neighborhood!



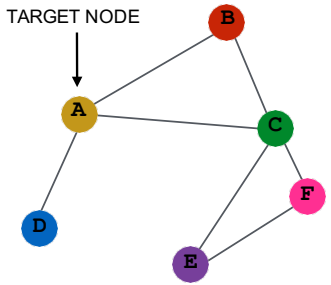
INPUT GRAPH



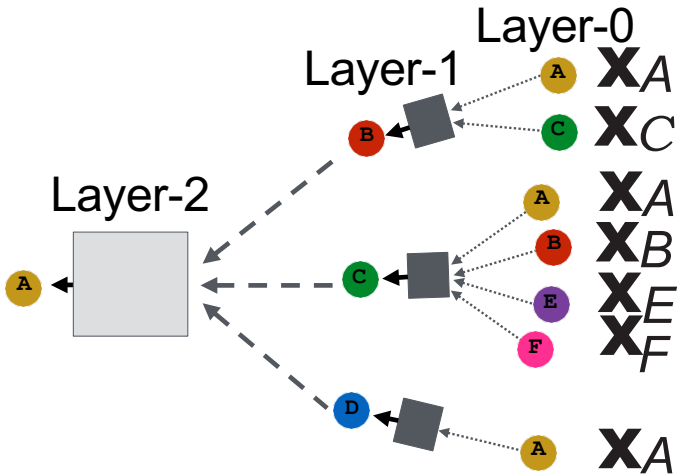
Deep Model: Many Layers

Model can be of arbitrary depth:

- Nodes have embeddings at each layer
- Layer-0 embedding of node v is its input feature, x_v
- Layer-k embedding gets information from nodes that are k hops away

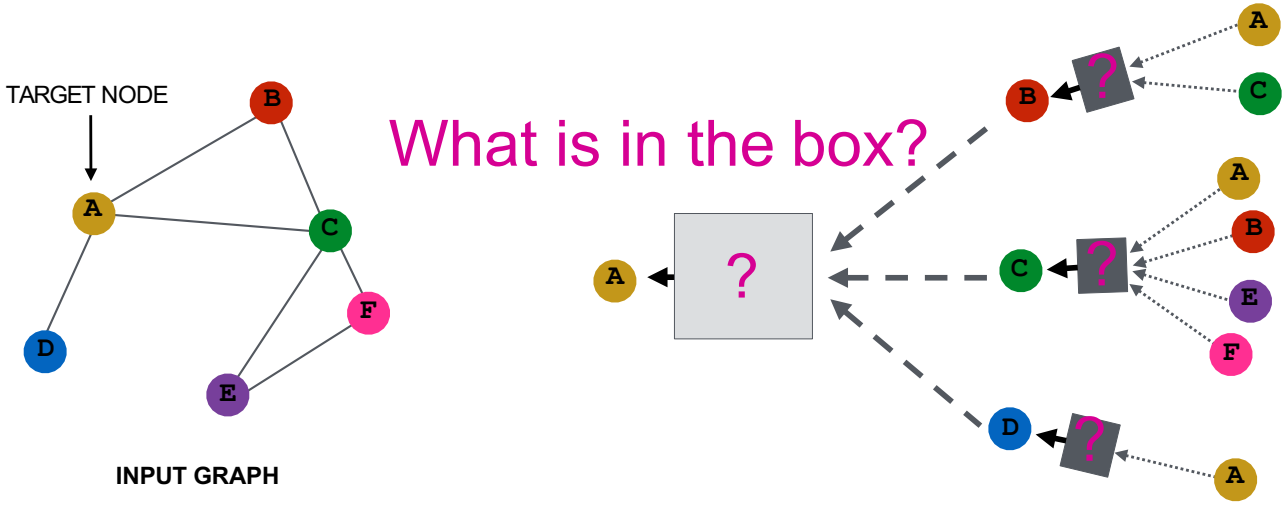


INPUT GRAPH



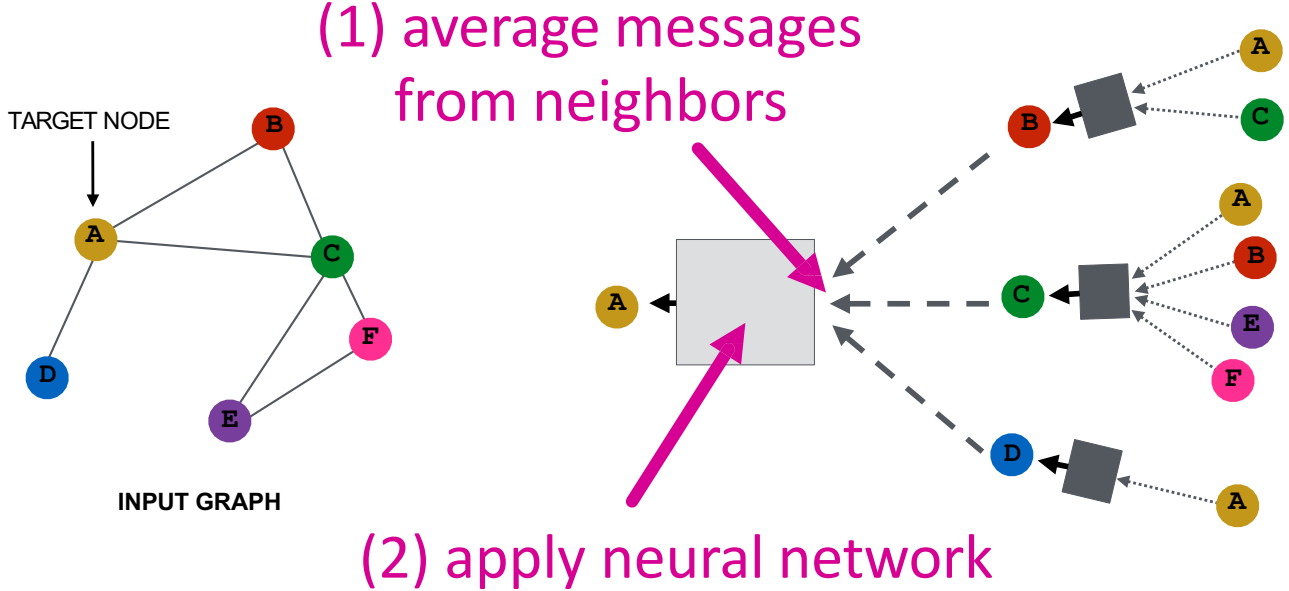
Neighborhood Aggregation

Neighborhood aggregation: Key distinctions are in how different approaches aggregate information across the layers



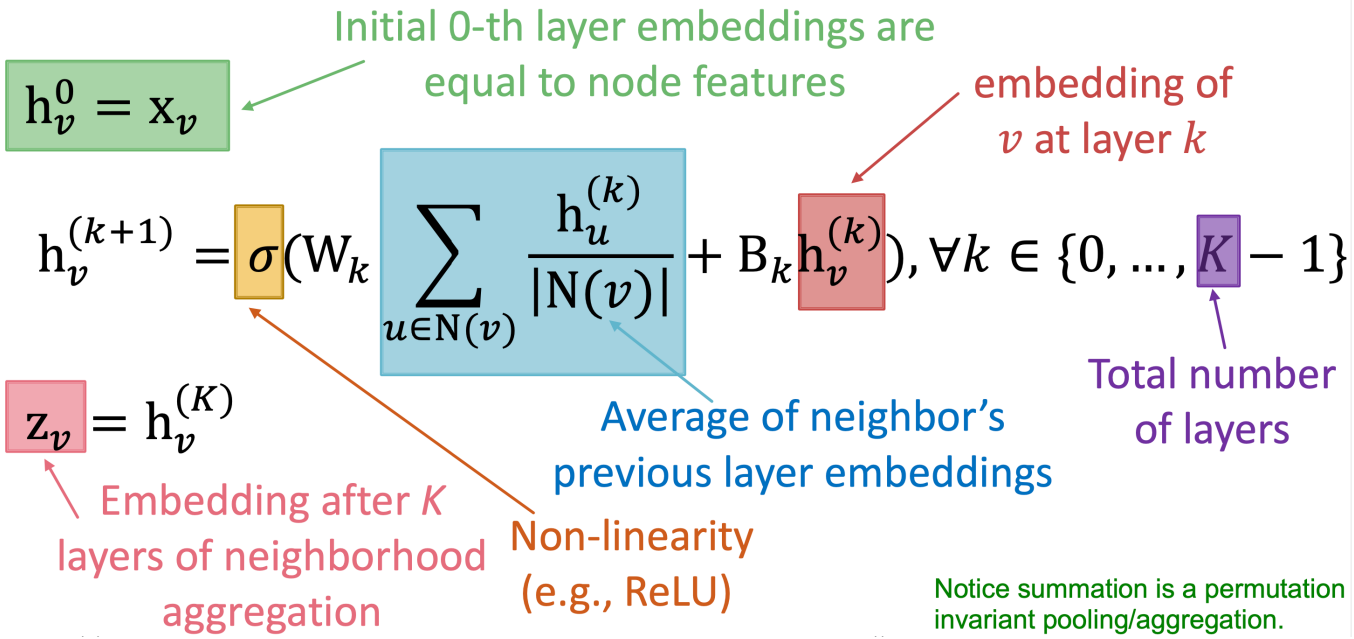
Neighborhood Aggregation

Basic approach: Average information from neighbors and apply a neural network



The Math: Deep Encoder

- **Basic approach:** Average neighbor messages and apply a neural network



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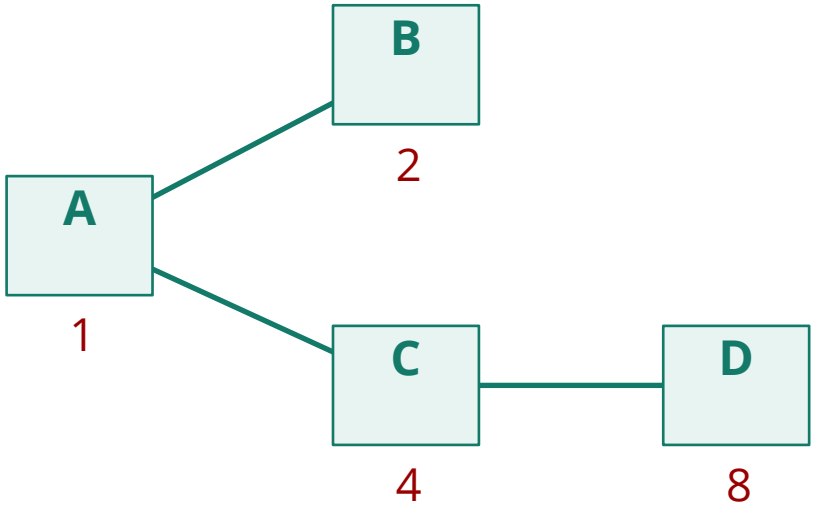
Read Only One-Hop Neighbors

All nodes: layer 0



A reads B and C.

Neighbor mean $= (2 + 4)/2 = 3$

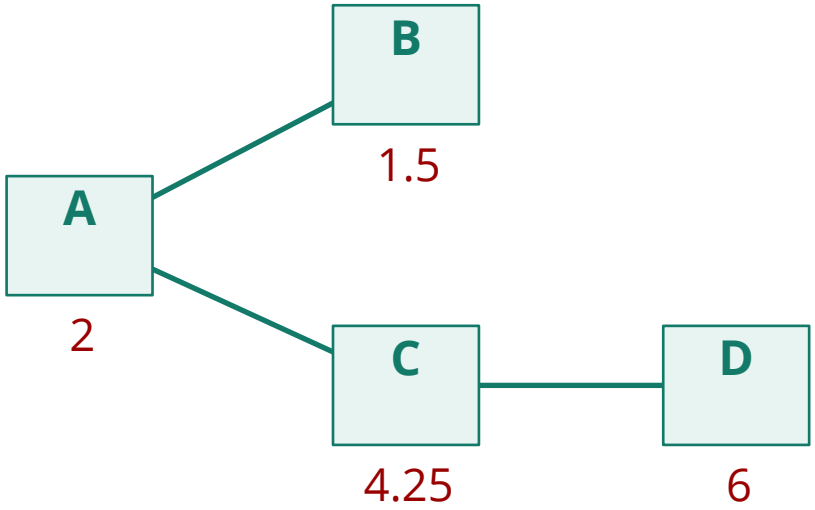
$A^1 = 0.5 \cdot 1 + 0.5 \cdot 3 = 2$

$h_v^{k+1} = 0.5\ h_v^k + 0.5\ \text{mean}\{h_u^k : u \text{ is a neighbor of } v\}$

Toy mean-aggregation update with fixed weights; not the exact normalized GCN rule.

Update Every Node From Layer 0

All nodes: layer 1



All layer-1 values:

$A^1 = 2, B^1 = 1.5$

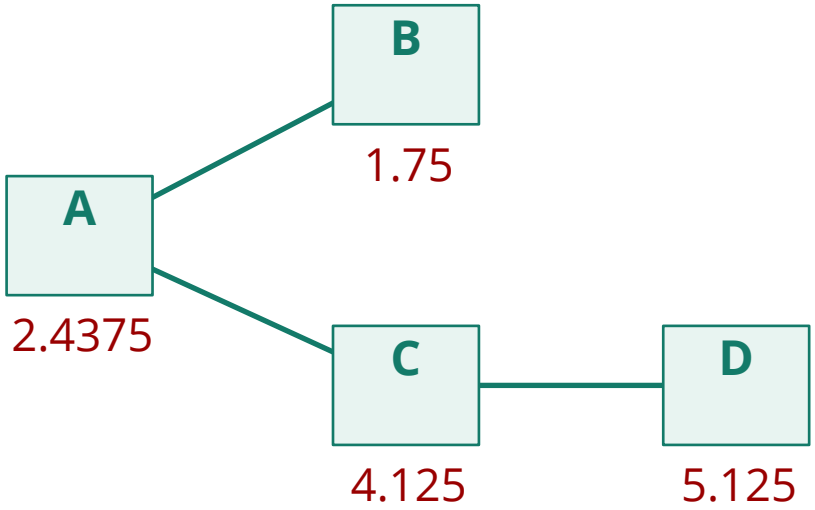
$C^1 = 4.25, D^1 = 6$

Every new value uses only the previous layer.

Toy mean-aggregation update with fixed weights; not the exact normalized GCN rule.

The Second Layer Reaches Two Hops

All nodes: layer 2



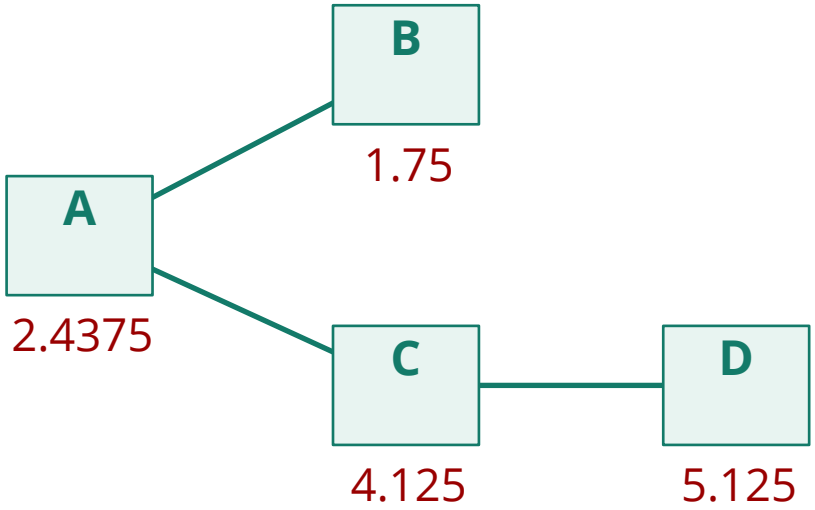
$$\text{A reads } B^1 \text{ and } C^1.$$
$$A^2 = 0.5 \cdot 2 + 0.5 \cdot (1.5 + 4.25)/2 = 2.4375$$

Two layers let information travel two graph edges.

Toy mean-aggregation update with fixed weights; not the exact normalized GCN rule.

Keep Updates Synchronous

All nodes: layer 2



Read layer k → write layer k+1

In-place A → B would give
$B^1 = 2$ instead of 1.5.

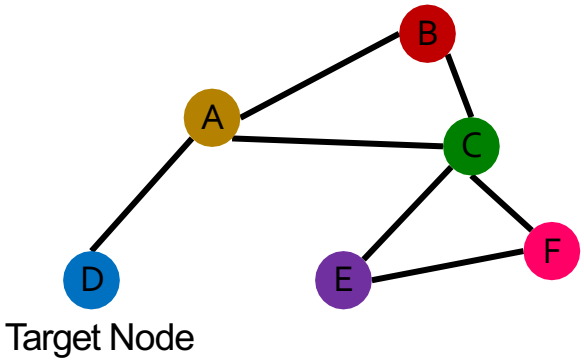
Node numbering does not define an update schedule.

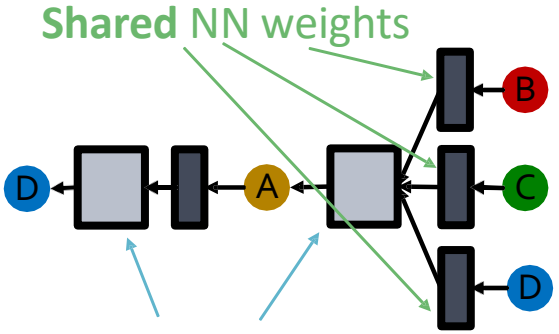
Toy mean-aggregation update with fixed weights; not the exact normalized GCN rule.

GCN: Invariance and Equivariance

What are the invariance and equivariance properties for a GCN?

Given a node, the GCN that computes its embedding is **permutation invariant**

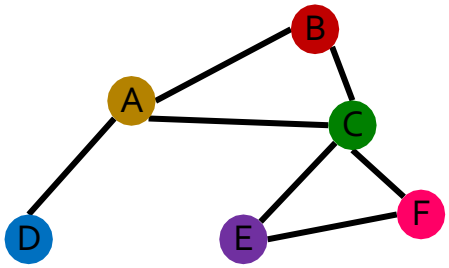




GCN: Invariance and Equivariance

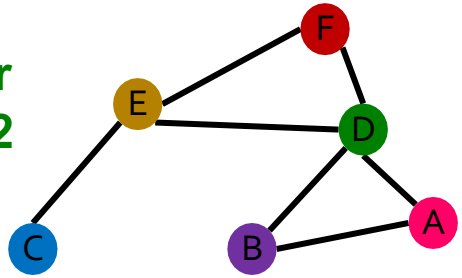
Considering all nodes in a graph, GCN computation is **permutation equivariant**

Order plan 1

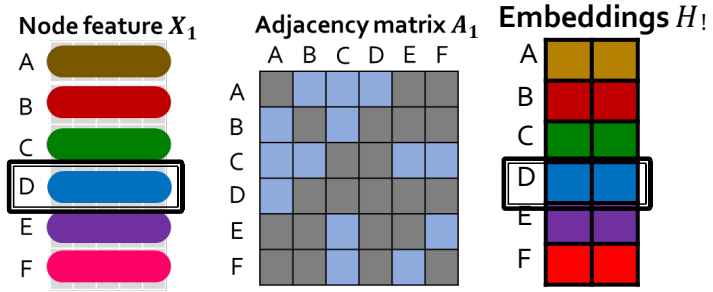


Target Node

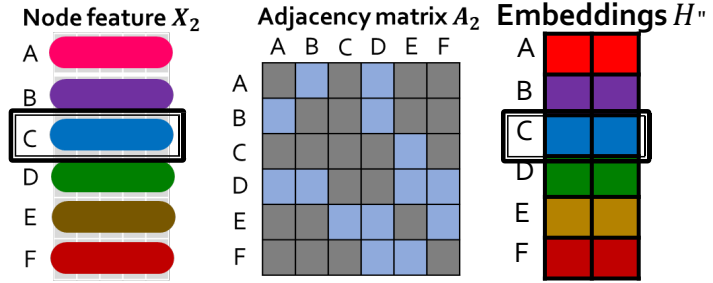
Order plan 2



Target Node



Permute the input, the output also permutes accordingly - permutation equivariant



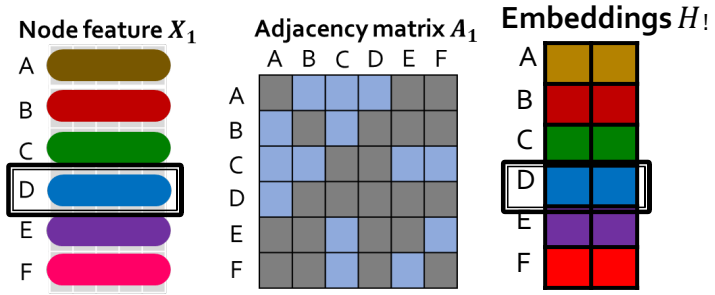
GCN: Invariance and Equivariance

Considering all nodes in a graph, GCN computation is **permutation equivariant**

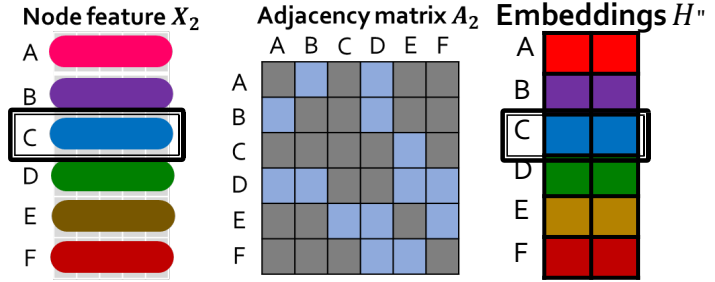
Detailed reasoning:

1. The rows of **input node features** and **output embeddings** are **aligned**
2. We know computing the embedding of **a given node** with GCN is **invariant.**
3. So, after permutation, the **location** of **a given node** in the **input node feature** matrix is changed, and the **the output embedding of a given node stays the same** (the colors of node feature and embedding are **matched**)

This is permutation equivariant

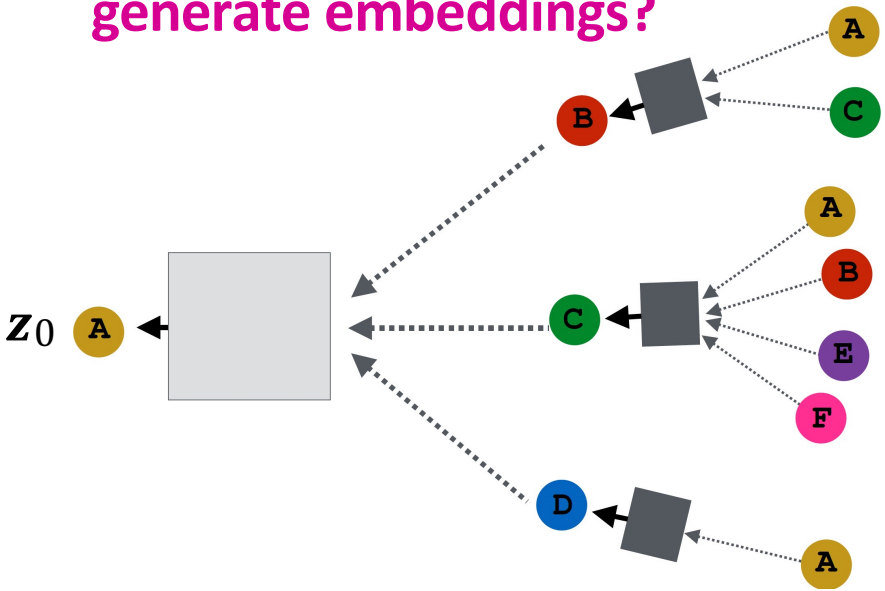


Permute the input, the output also permutes accordingly - permutation equivariant



Training the Model

How do we train the GCN to generate embeddings?



Need to define a loss function on the embeddings.

Model Parameters

Trainable weight matrices
(i.e., what we learn)

$$h_v^{(0)} = x_v$$

$$h_v^{(k+1)} = \sigma\left(W_k \sum_{u \in N(v)} \frac{h_u^{(k)}}{|N(v)|} + B_k h_v^{(k)}\right), \forall k \in \{0..K-1\}$$

$$z_v = h_v^{(K)}$$

Final node embedding

We can feed these **embeddings into any loss function** and run SGD to **train the weight parameters**

h_v^k: the hidden representation of node v at layer k

- W_k: weight matrix for neighborhood aggregation
- B_k: weight matrix for transforming hidden vector of self

How to Train A GNN

Node embedding $\boldsymbol{z}_0$ is a function of input graph

Supervised setting: We want to minimize loss $\mathcal{L}$:

$$\min_{\Theta} \mathcal{L}(\boldsymbol{y}, f_{\Theta}(\boldsymbol{z}_v))$$

- $\boldsymbol{y}$: node label
- $\mathcal{L}$ could be L2 if $\boldsymbol{y}$ is real number, or cross entropy if $\boldsymbol{y}$ is categorical

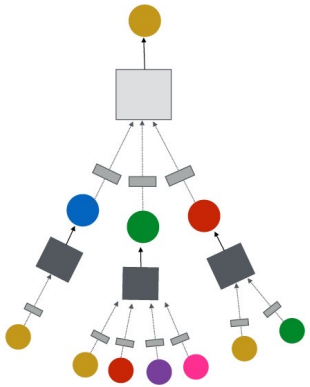
Unsupervised setting:

- No node label available
- **Use the graph structure as the supervision!**

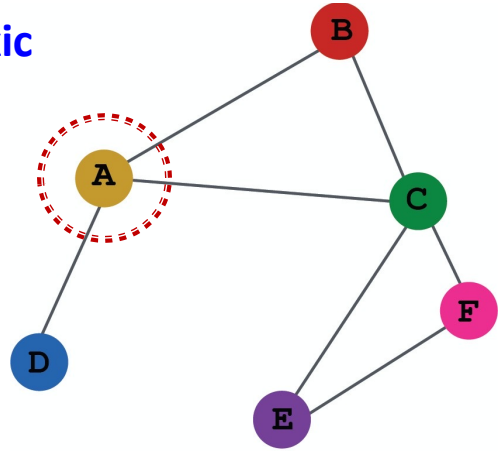
Supervised Training

Directly train the model for a supervised task (e.g., **node classification**)

Safe or toxic drug?



Safe or toxic drug?



E.g., a drug-drug interaction network

Supervised Training

Directly train the model for a supervised task (e.g., **node classification**)

- Use cross entropy loss

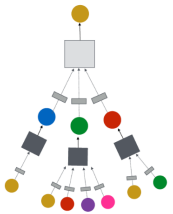
$$\mathcal{L} = -\sum_{v \in V} y_v \log(\sigma(z_v^T \theta)) + (1 - y_v)\log(1 - \sigma(z_v^T \theta))$$

Encoder output: node embedding

Classification weights

Node class label

Safe or toxic drug?



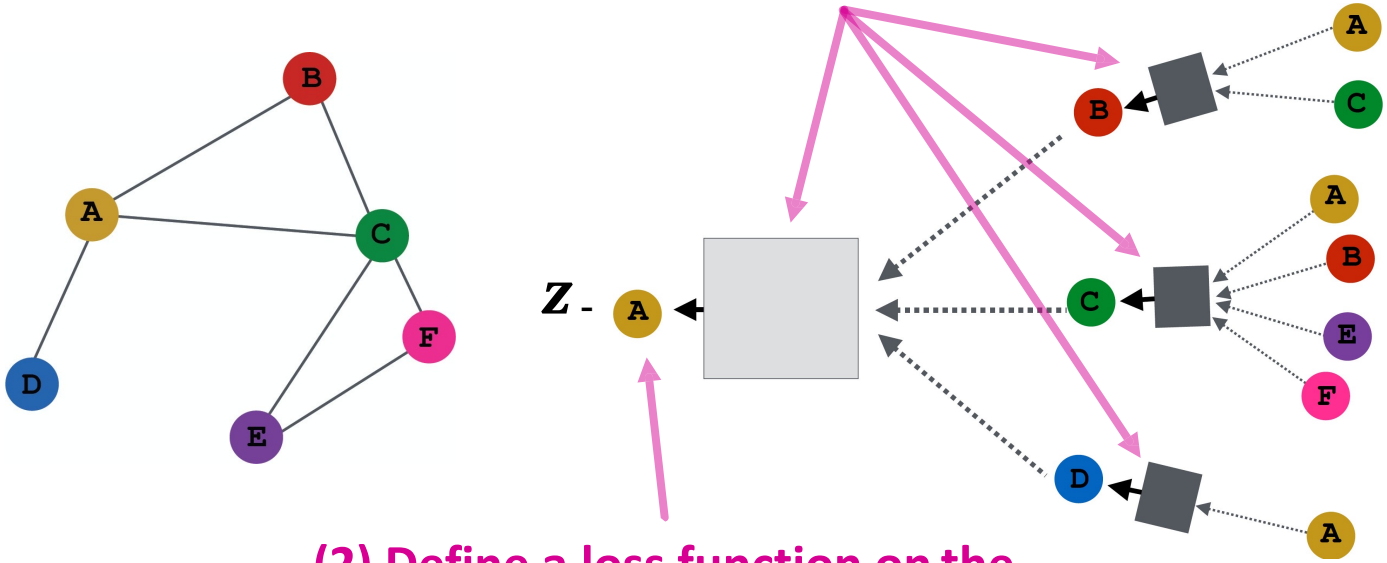
True probability distribution (one-shot)

$$H(p, q) = -\sum_{x \in classes} p(x) \log q(x)$$

Your model's predicted probability distribution

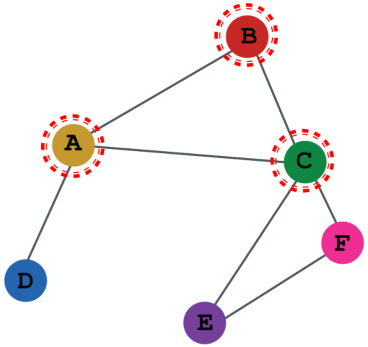
Model Design: Overview

(1) Define a neighborhood aggregation function



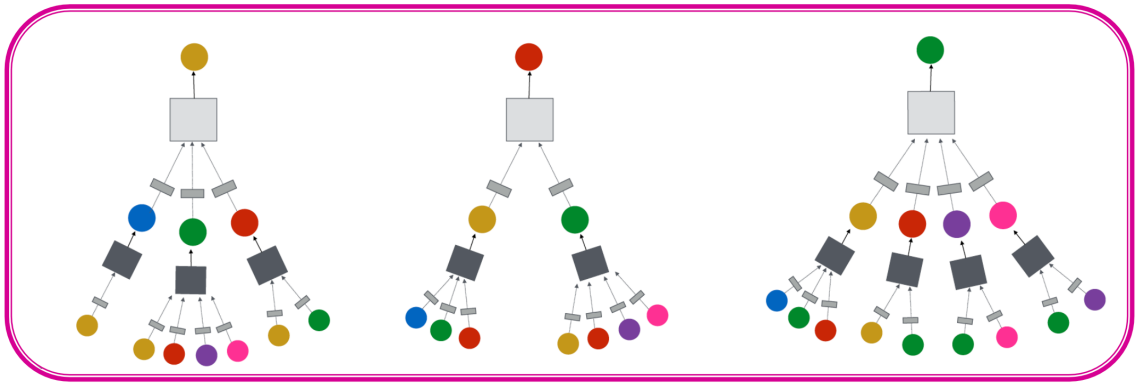
(2) Define a loss function on the embeddings

Model Design: Overview

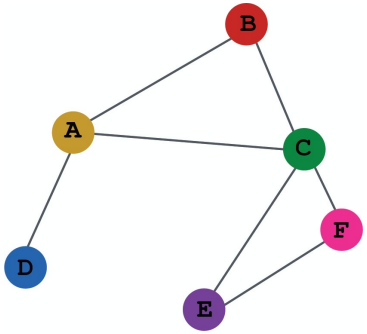


INPUT GRAPH

(3) Train on a set of nodes, i.e., a batch of compute graphs



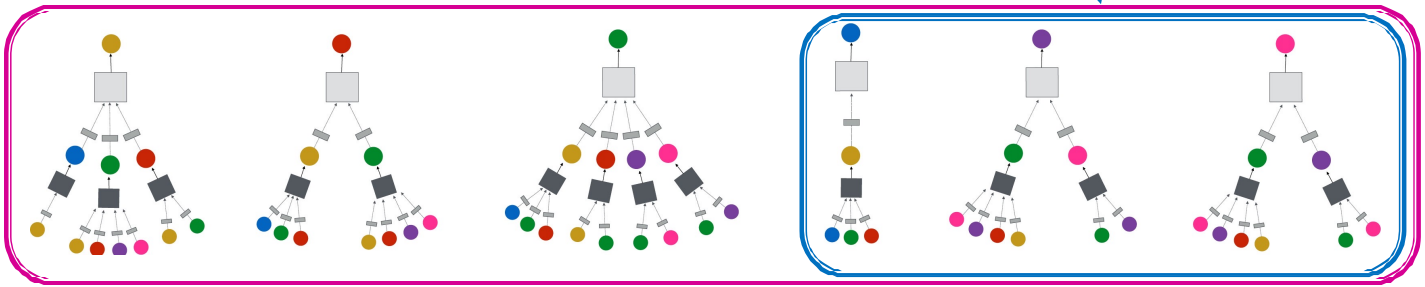
Model Design: Overview



INPUT GRAPH

(4) Generate embeddings for nodes as needed

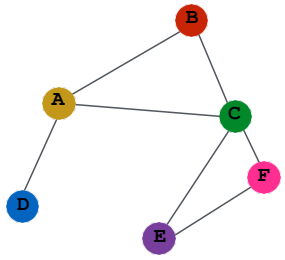
Even for nodes we never trained on!



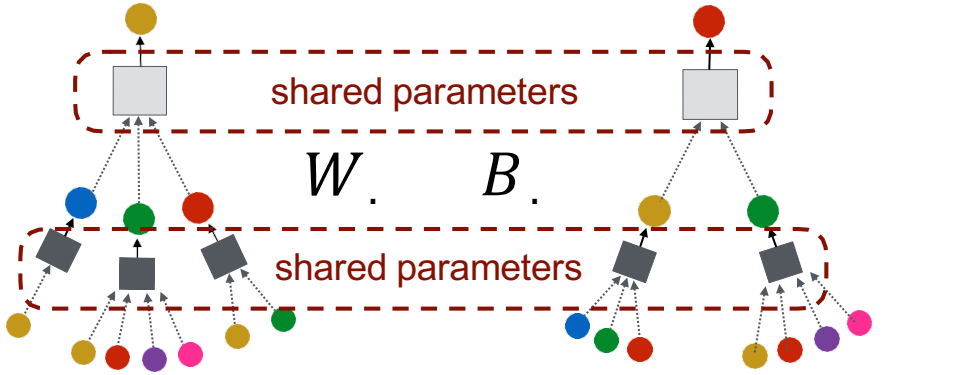
Inductive Capability

The same aggregation parameters are shared for all nodes:

- Parameter count does not depend on $|V|$ (fixed width/depth); reuse on new nodes.



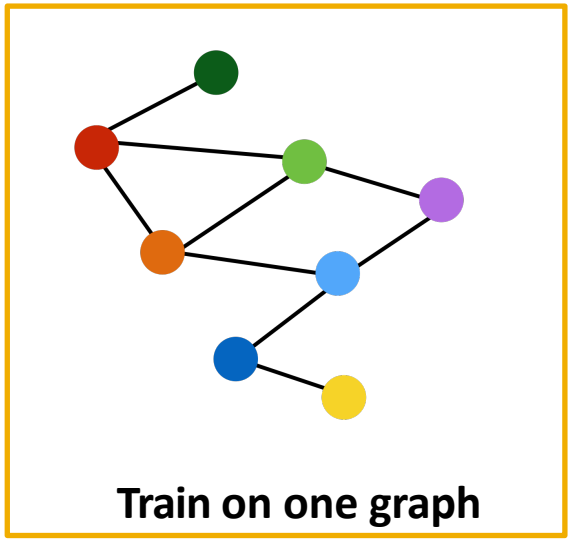
INPUT GRAPH

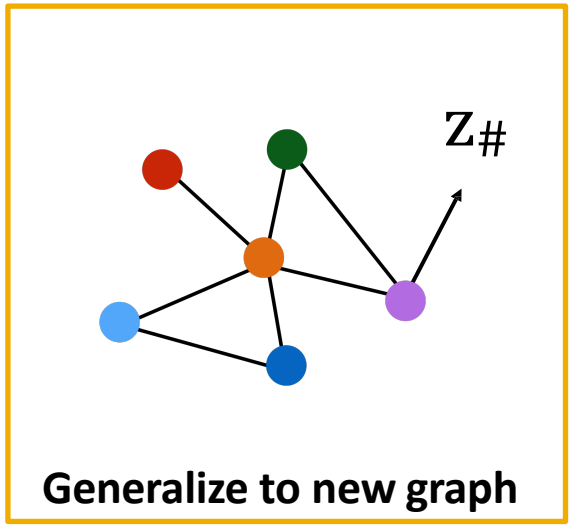


Compute graph for node A

Compute graph for node B

Inductive Capability: New Graphs

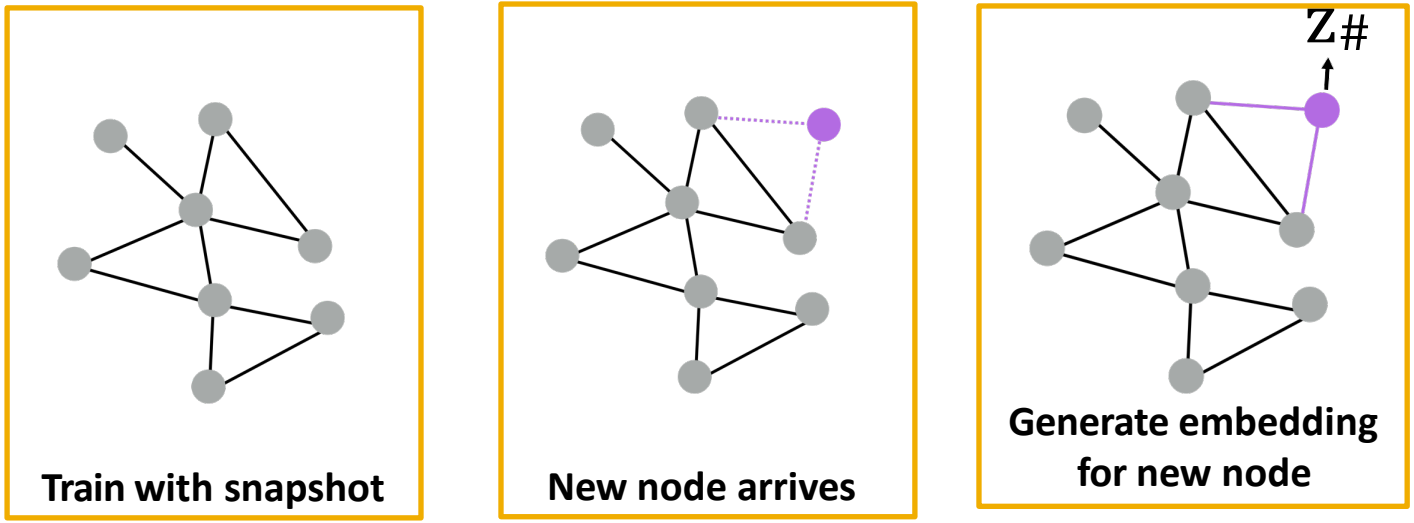




Inductive node embedding → Generalize to entirely unseen graphs

E.g., train on protein interaction graph from model organism A and generate embeddings on newly collected data about organism B

Inductive Capability: New Nodes



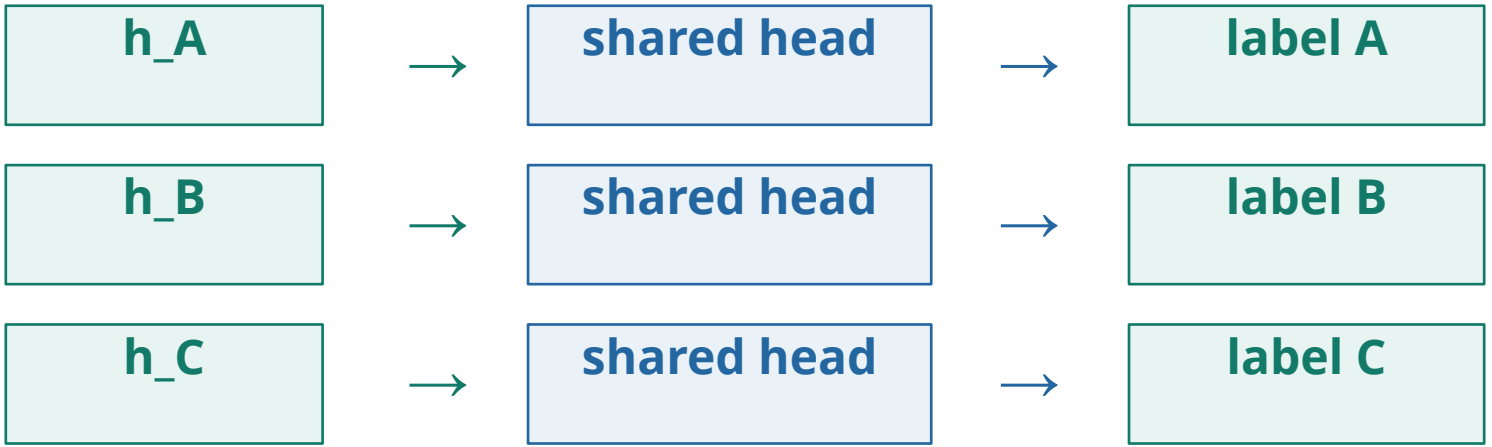
Many application settings constantly encounter previously unseen nodes:

- E.g., Reddit, YouTube, Google Scholar

Need to generate new embeddings "on the fly"

Node Outputs Keep Their Identity

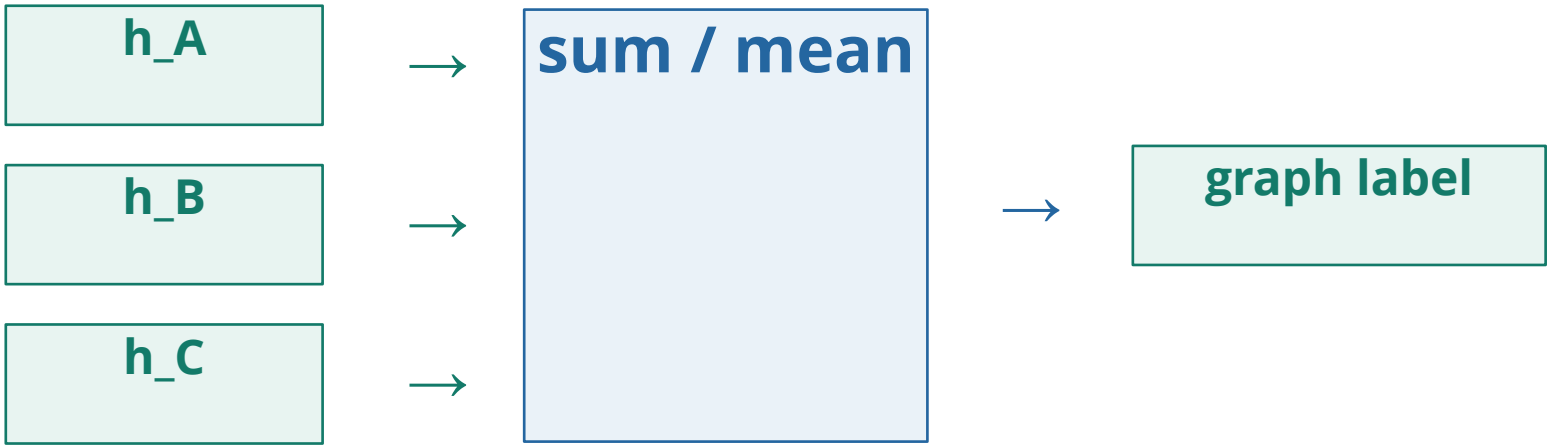
1 / 2



Reordering node inputs reorders the corresponding node outputs.

Course schematic.

Graph Outputs Pool Across Nodes



A symmetric readout gives the same graph output after reordering.

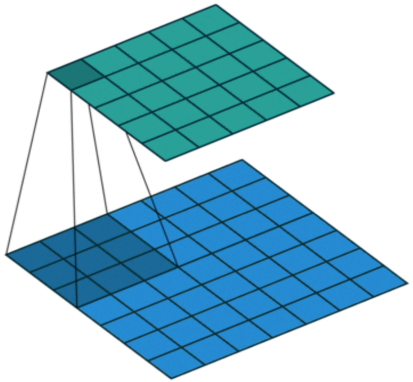
Course schematic.

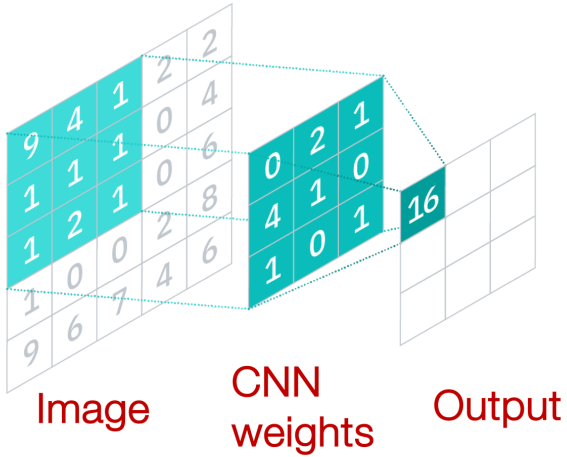
Architecture Comparison

How do GNNs compare to prominent architectures such as Convolutional Neural Nets?

Convolutional Neural Network

Convolutional neural network (CNN) layer with 3x3 filter:



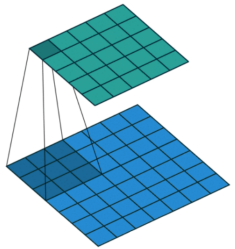


CNN formulation: $\mathrm{h}_v^{(l+1)} = \sigma(\sum_{u \in \mathrm{N}(v) \cup \{v\}} \mathrm{W}_l^u \mathrm{h}_u^{(l)}), \quad \forall l \in \{0, \ldots, L-1\}$

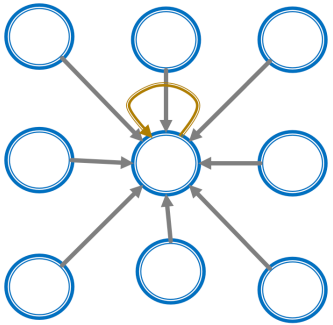
$N(v)$ represents the 8 neighbor pixels of v.

GNN vs. CNN

Convolutional neural network (CNN) layer with 3x3 filter:



Image



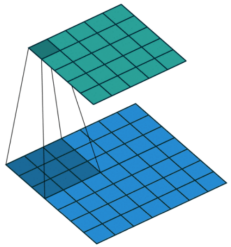
Graph

- GNN formulation: $\mathrm{h}_v^{(l+1)} = \sigma(\mathbf{W}_l \sum_{u \in \mathrm{N}(v)} \frac{\mathrm{h}_u^{(l)}}{|\mathrm{N}(v)|} + \mathrm{B}_l \mathrm{h}_v^{(l)}), \forall l \in \{0, \ldots, L-1\}$
- CNN formulation: (previous slide) $\mathrm{h}_v^{(l+1)} = \sigma(\sum_{u \in \mathrm{N}(v) \cup \{v\}} \mathrm{W}_l^u \mathrm{h}_u^{(l)}), \forall l \in \{0, \ldots, L-1\}$

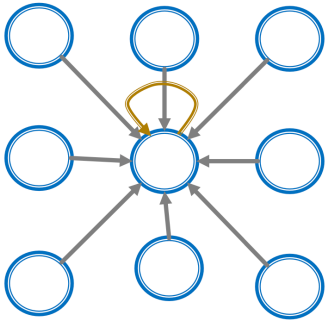
  if we rewrite: $\mathrm{h}_v^{(l+1)} = \sigma(\sum_{u \in \mathrm{N}(v)} \mathbf{W}_l^u \mathrm{h}_u^{(l)} + \mathrm{B}_l \mathrm{h}_v^{(l)}), \forall l \in \{0, \ldots, L-1\}$

GNN vs. CNN

Convolutional neural network (CNN) layer with 3x3 filter:



Image



Graph

GNN formulation: $\mathrm{h}_v^{(l+1)} = \sigma(\mathbf{W}_l \sum_{u \in \mathrm{N}(v)} \frac{\mathrm{h}_u^{(l)}}{|\mathrm{N}(v)|} + \mathrm{B}_l \mathrm{h}_v^{(l)}), \forall l \in \{0, \ldots, L-1\}$

CNN formulation: $\mathrm{h}_v^{(l+1)} = \sigma(\sum_{u \in \mathrm{N}(v)} \mathbf{W}_l^u \mathrm{h}_u^{(l)} + \mathrm{B}_l \mathrm{h}_v^{(l)}), \forall l \in \{0, \ldots, L-1\}$

Key difference: We can learn different W_l^u for different "neighbor" u for pixel v on the image. The reason is we can pick an order for the 9 neighbors using **relative position** to the center pixel: {(-1,-1). (-1,0), (-1, 1), …, (1, 1)}

GNN vs. CNN

Convolutional neural network (CNN) layer with 3x3 filter:

- **CNN can be seen as a special GNN with fixed neighbor size and ordering:**
  - **The size of the filter is pre-defined for a CNN.**
  - **The advantage of GNN is it processes arbitrary graphs with different degrees for each node.**
- **CNNs use fixed spatial offsets on a grid.**
  - **Arbitrary pixel permutations do not preserve these offsets.**

Key difference: We can learn different W^{μ} for different "neighbor" u for pixel v on the image. The reason is we can pick an order for the 9 neighbors using **relative position** to the center pixel: {(-1,-1). (-1,0), (-1, 1), …, (1, 1)}

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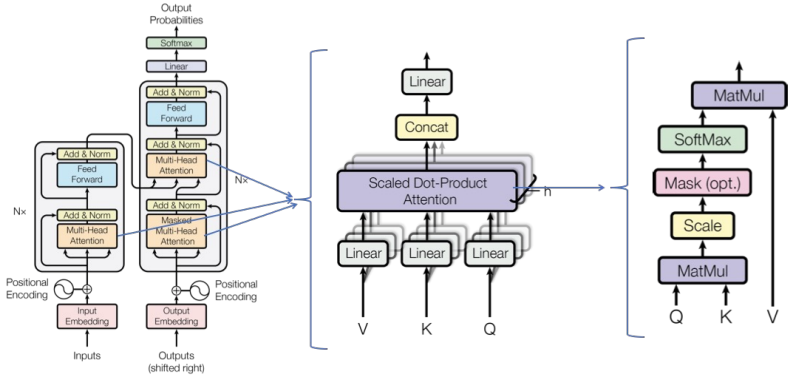
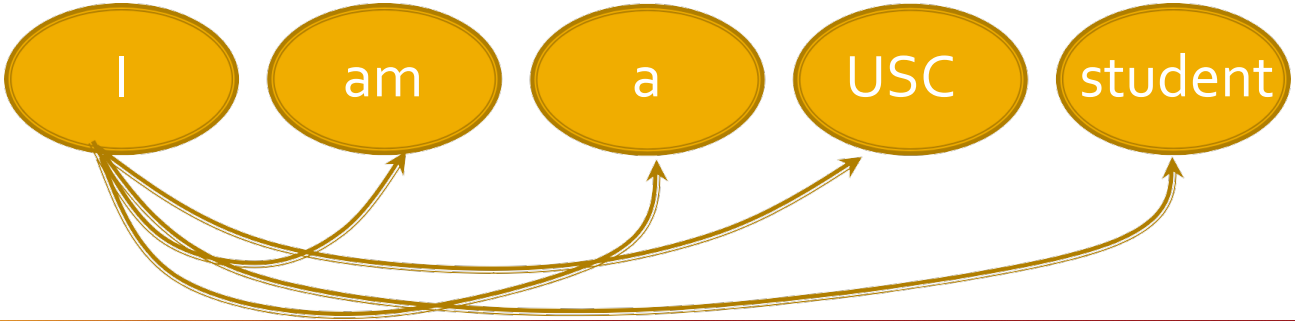
Transformer

Transformer is one of the most popular architectures that achieves great performance in many sequence modeling tasks.



Key component: self-attention

Every token/word attends to all the other tokens/words via matrix calculation.



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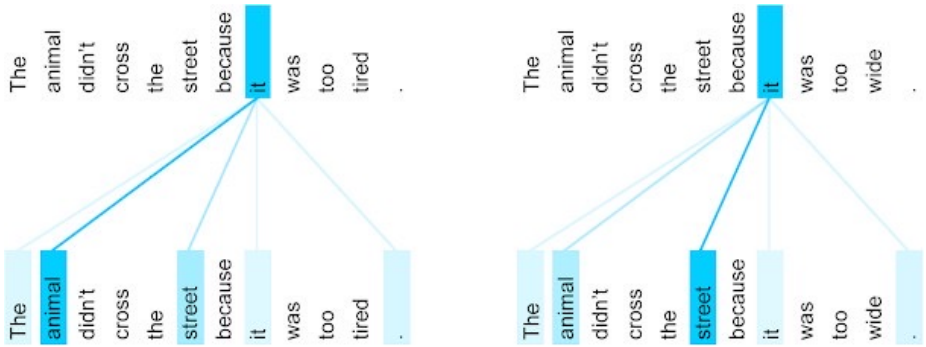
Transformer

A general definition of attention:

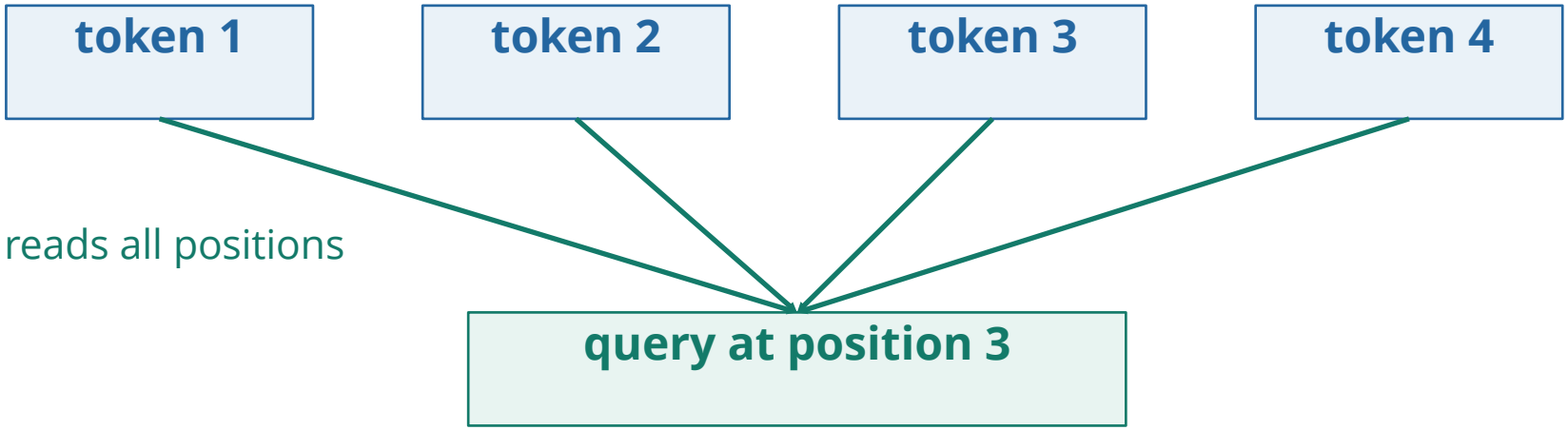
Given a set of vector *values*, and a vector *query*, **attention** is a technique to compute a weighted sum of the values, dependent on the query.

Each token/word has a **value vector** and a **query vector**. The value vector can be seen as the representation of the token/word. We use the query vector to calculate the attention score (weights in the weighted sum).



Attention Defines Allowed Edges

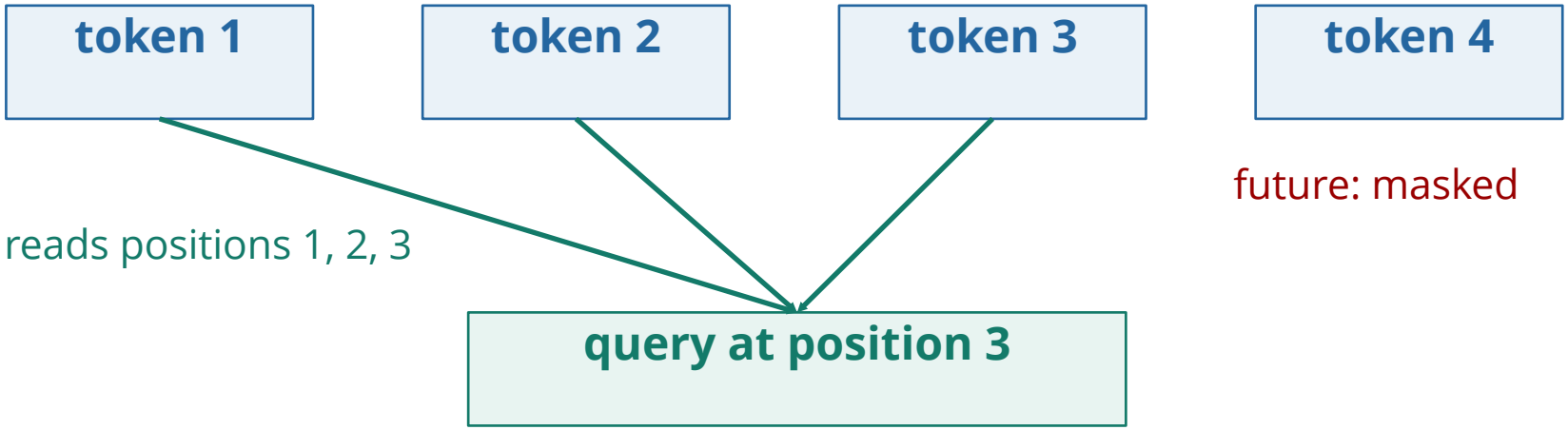
1 / 2



Unmasked self-attention uses a complete token interaction graph.

Course schematic.

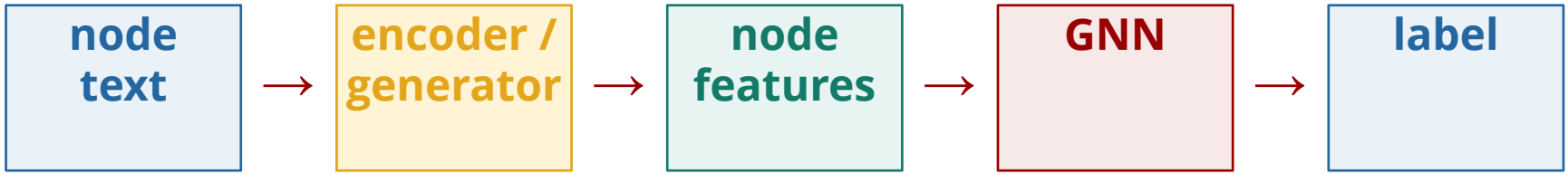
Causal Attention Removes Future Edges



The causal mask makes the allowed interactions directional.

Course schematic.

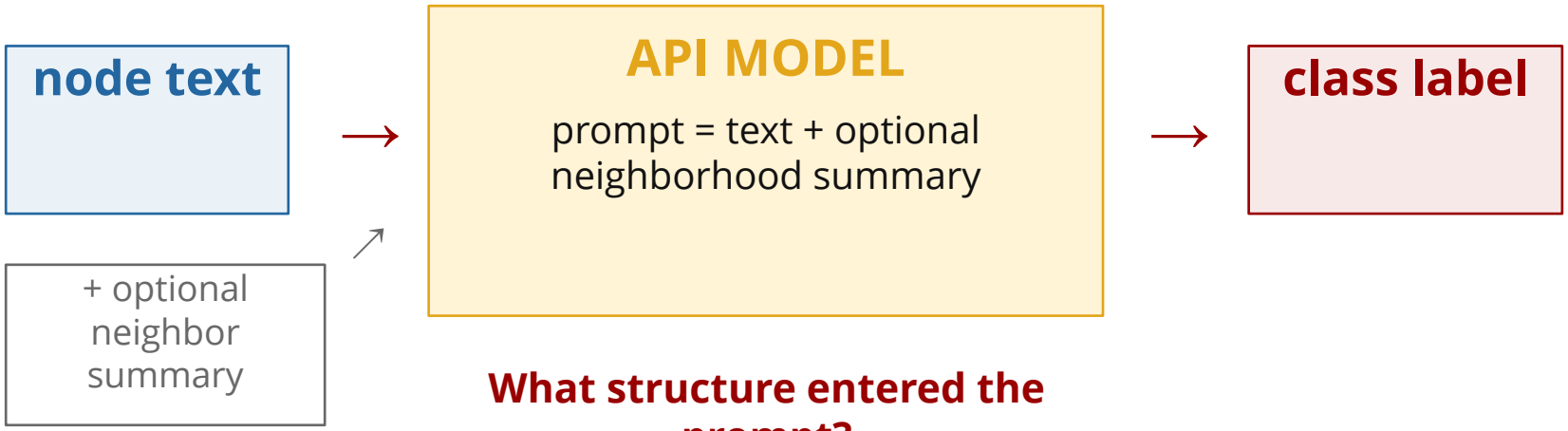
The 2023 Baseline: Enhance Then Predict



language model changes the inputs; the GNN makes the prediction

Pipeline 1: semantics become graph features.

The 2023 Baseline: Predict Directly



What structure entered the prompt?

Pipeline 2: direct labels do not prove graph reasoning.

The 2026 Map Has Four System Families

FEATURES

How does text enter a GNN?

FOUNDATION

What transfers to unseen graphs?

GraphRAG

Which evidence subgraph is retrieved?

TOOL AGENT

Which graph operation runs next?

Name the family before comparing results.

Four Families, Four Failure Surfaces

FEATURES

failure: leakage or weak edges

FOUNDATION

failure: transfer breaks across tasks

GraphRAG

failure: noisy graph or missed evidence

TOOL AGENT

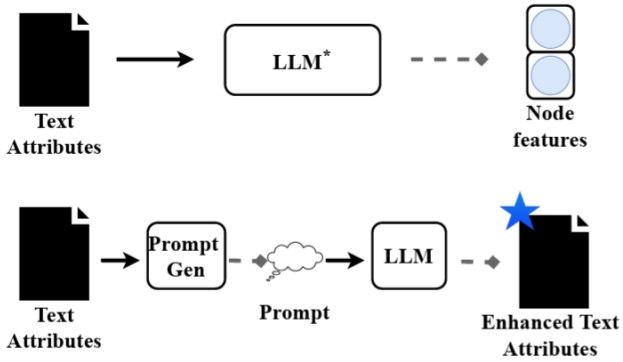
failure: wrong tool or wrong sequence

A final answer score cannot locate the failed stage.

Pipelines for LLMs & Graphs

1. LLMs-as-Enhancers

- LLM → enhance/encode node text → GNN → final prediction.
- Feature-level or text-level augmentation.

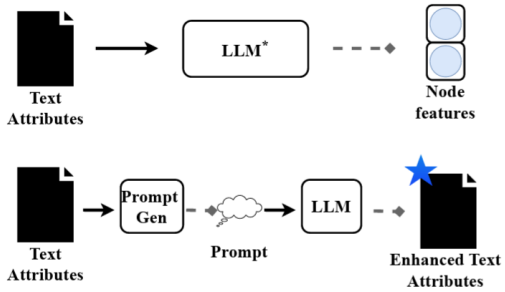


(a) *An illustration of LLMs-as-Enhancers*, where LLMs preprocess the text attributes, and GNNs eventually make the predictions. Three different structures for this pipeline are demonstrated in Figure 2.

LLMs as Enhancers

Feature-Level

- **Cascading:** Sentence-BERT or e5 → feed embeddings to GNN.
  - Often very effective and efficient.
- **Iterative:** GNN & PLM exchange pseudo-labels.
  - Can be powerful if enough labeled data.



(a) *An illustration of LLMs-as-Enhancers*, where LLMs preprocess the text attributes, and GNNs eventually make the predictions. Three different structures for this pipeline are demonstrated in Figure 2.

Text-Level

- **TAPE [22]:** LLM writes pseudo labels + explanations → re-encoded by PLMs → GNN.
- **KEA:** LLM adds knowledge entities to node text → encode + ensemble.

LLMs as Enhancers

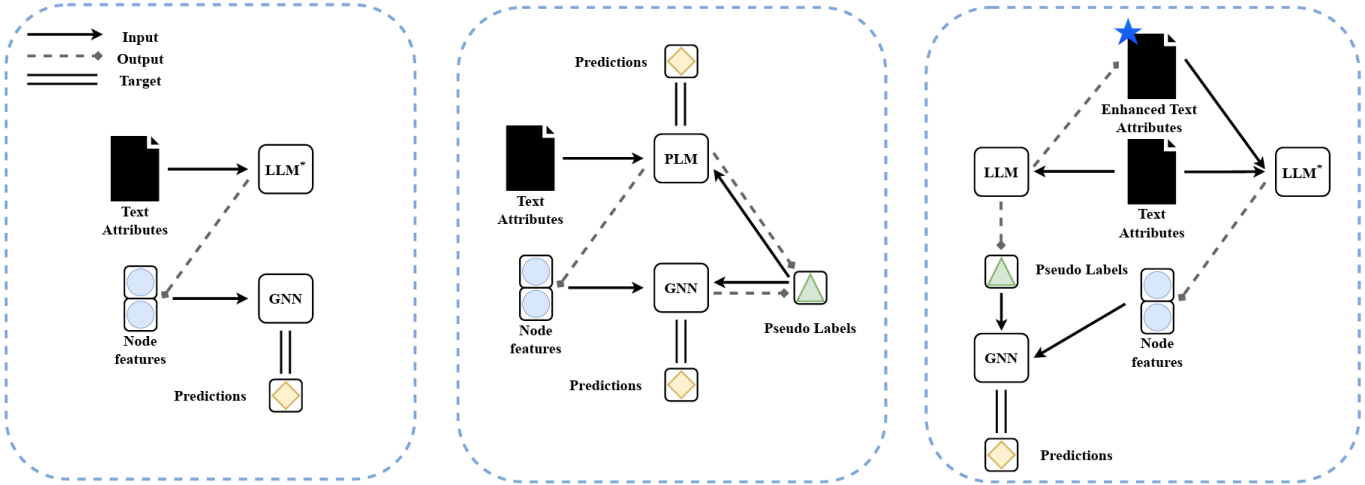
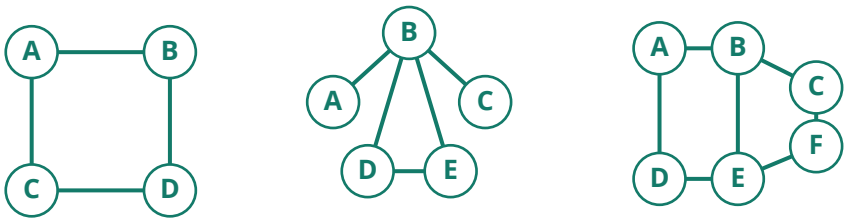


Figure 2: Three strategies to adopt LLMs as enhancers. The first two integrating structures are designed for feature-level enhancement, while the last structure is designed for text-level enhancement. From left to right: (1) Cascading Structure: Embedding-visible LLMs enhance text attributes directly by encoding them into initial node features for GNNs. (2) Iterative Structure: GNNs and PLMs are co-trained in an iterative manner. (3) Text-level enhancement structure: Embedding-invisible LLMs are initially adopted to enhance the text attributes by generating augmented attributes. The augmented attributes and original attributes are encoded and then ensembled together.

Graph Foundation Models Transfer Structure

FEATURES → FOUNDATION



PRETRAIN ACROSS DIFFERENT GRAPHS

→

TRANSFER RULES

node equivariance
label equivariance
feature invariance

↓

UNSEEN GRAPH OR TASK

Transfer requires the right symmetries, not just more parameters.

LLM as the Graph-Task Predictor

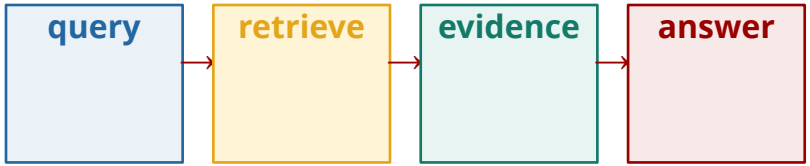
Enhancer: LLM-derived features feed a separate graph predictor.
Predictor: the LLM supplies the final task prediction.

Compare text-only and graph-context inputs under the same split.

Chen et al. (2023), Exploring the Potential of LLMs in Learning on Graphs. Teaching schematic.

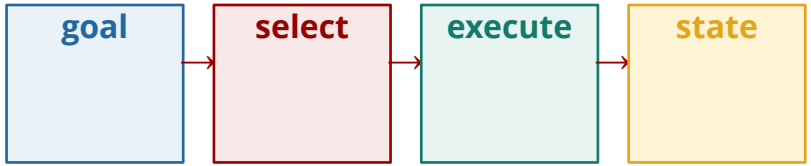
GraphRAG Retrieves; Graph Agents Orchestrate

GraphRAG



retrieve evidence

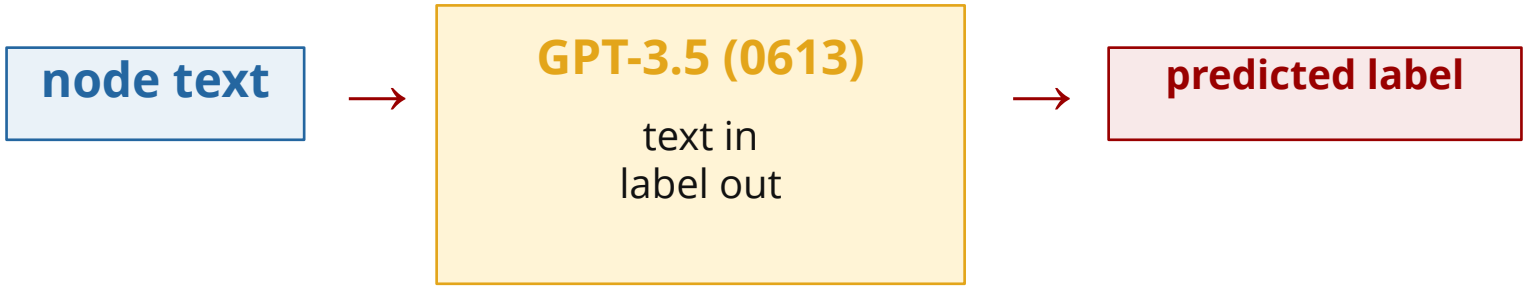
Tool agent



orchestrate operations

Keep the large graph outside the LM context.

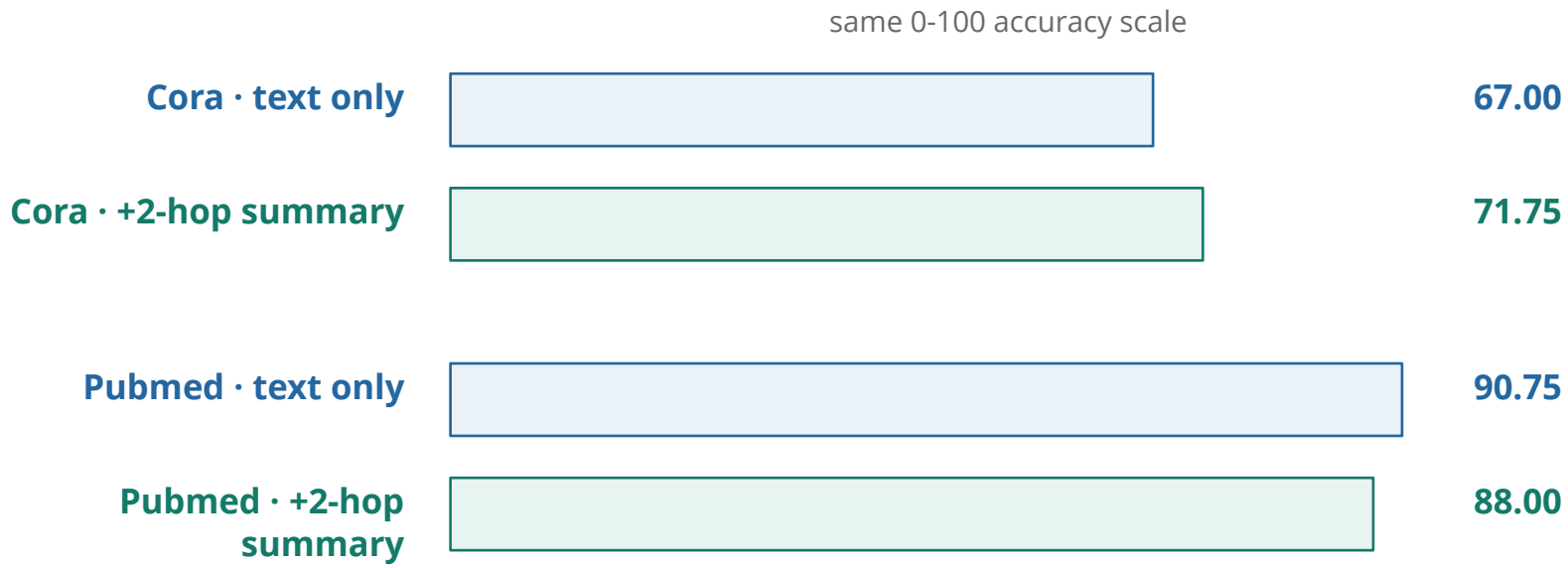
What Did the Source Result Actually Test?



HEADLINE ✓	SEPARATE TABLE ✓	TOOLS ✗	RANDOM LABELS ✗
node text only	+ 2-hop summary	not used	not tested

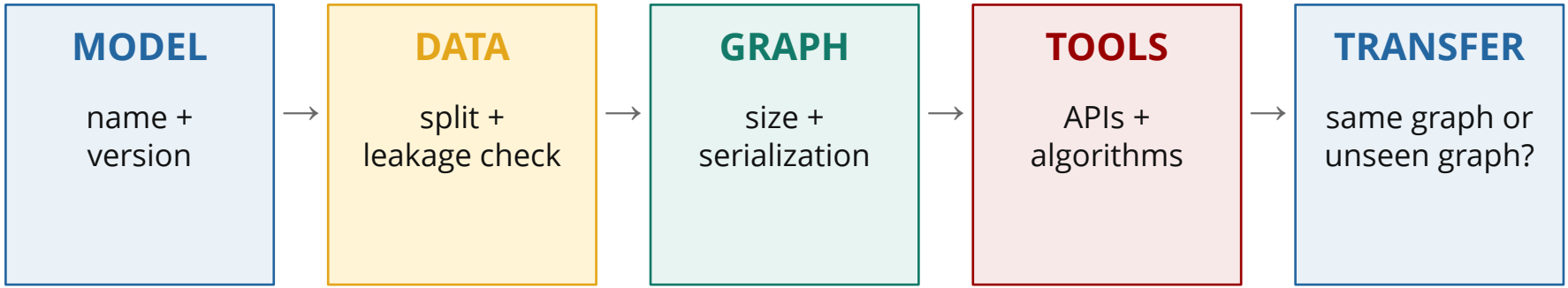
The headline score is text classification evidence.

Neighborhood Evidence Changes Results



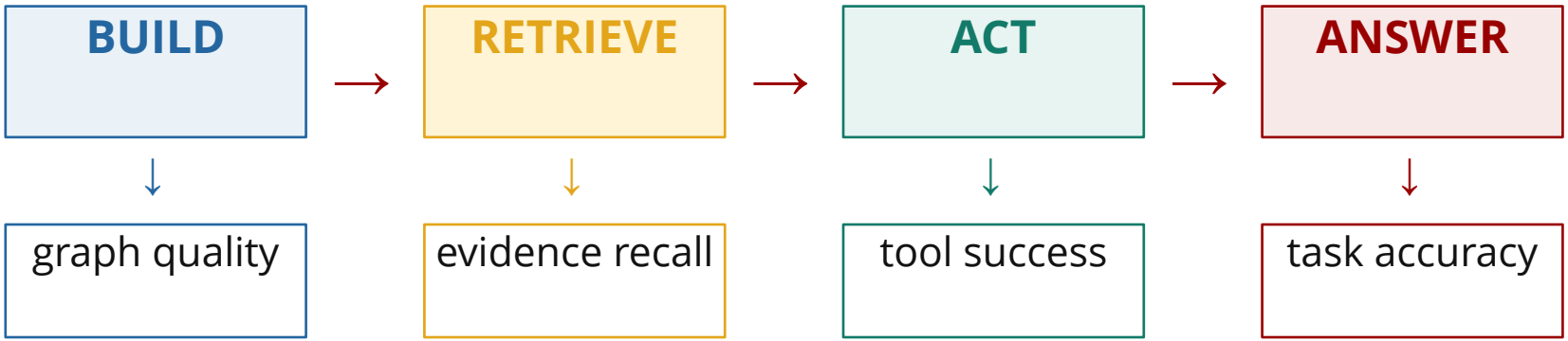
Neighborhood summaries change results in mixed directions.

Stress-Test an “LLM for Graphs” Claim



Report the setup before reporting the score.

Score the System Where It Can Fail



Evaluate each stage that the chosen system actually uses.

Summary

In this lecture, we introduced

- Idea for Deep Learning for Graphs
  - Multiple layers of embedding transformation
  - At every layer, use the embedding at previous layer as the input
  - Aggregation of neighbors and self-embeddings
- Graph Convolutional Network
  - Mean aggregation (can be expressed in matrix form)
- GNN is a general architecture
  - CNN can be viewed as a special GNN
- GNN + LLMs – new directions to follow!